Cross-Dataset Reliability and Computational Trade-offs in Cascaded URL Phishing Detection

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Dedication

I want to express my appreciation to my family and everyone who has supported me along the way. I would like to thank the countless mentors, teachers and managers throughout my career who supported me – even though they didn't see everything I had accomplished at that time. They saw the potential I had, and they believed I would succeed. I value all those experiences to this day. I also wish to recognize those people and organizations that have worked behind the scenes to make sure our critical systems continue to operate smoothly and safely.

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Abstract of Praxis

Cross-Dataset Reliability and Computational Trade-offs in Cascaded URL Phishing Detection

This Praxis builds and tests CyberSentinel, a uniform resource locator (URL) phishing gateway combining structural classification, a character transformer, distribution monitoring and a Hypertext Transfer Protocol (HTTP) service. Internal structural recall reaches 98.6807% at a 0.7747% false-positive rate, a 64.07-percentage-point gain over length alone. Controlled experiments explain the system's behavior and test improvements. Validation labels are strongly associated with URL format, and scheme spelling changes frozen-model decisions; these tests do not quantify the format's share of the internal gain. On the evaluated external cohort, shared errors impose an exact 90.5487% false-positive floor on routing between unchanged decisions. Monitoring recognizes 87.12% of external windows, exceeding its sensitivity target, while audit false alerts exceed their limit. Escalation adds 218 false positives without another gold detection, establishing the need to evaluate alerts and subsequent decisions separately. In single-host synthetic structural-only service tests, worker-owned clients reduce median paired 95th-percentile latency by 79.68%, preserving every comparable prediction. A no-model control supports the client-path explanation. The complete gateway did not meet its prespecified joint hypotheses; the experiments identify the governing constraints and test concrete remedies. CyberSentinel contributes a working architecture, a measured service improvement, a benchmark-format check and an exact bound that guides expert selection.

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List of Symbols

TP, FP, TN, FN — True-positive, false-positive, true-negative and false-negative counts against source reference labels.

P, N — Positive and negative stratum sizes in confusion-count tables; a total stream size is identified explicitly where used.

X; P(X) — Observed feature vector; its marginal distribution.

τ — Validation-selected decision threshold, unchanged during evaluation.

p50, p95, p99 — 50th, 95th and 99th percentiles of individual measured terminal request latencies.

pp — Percentage points; the unit of differences between proportions expressed as percentages.

c — HTTP client concurrency, such as c64 for 64 concurrent clients.

List of Acronyms

AC — Alternating current; the monitored external-power condition.

AI — Artificial intelligence.

AP — Average precision.

API — Application programming interface.

ASCII — American Standard Code for Information Interchange.

ASGI — Asynchronous Server Gateway Interface.

AUC — Area under the curve.

BIC — Bayesian information criterion.

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CI — Continuous integration in the software-verification record; uncertainty intervals are written out in the manuscript.

CLI — Command-line interface.

CP — Clopper–Pearson.

CPU — Central processing unit.

CSV — Comma-separated values.

DNS — Domain Name System.

DOI — Digital object identifier.

ECE — Expected calibration error.

FIFO — First-in, first-out queue discipline.

FNR — False-negative rate.

FPR — False-positive rate.

GB — Gigabytes.

GELU — Gaussian error linear unit.

GMM — Gaussian mixture model.

HTTP — Hypertext Transfer Protocol.

HTTPS — Hypertext Transfer Protocol Secure.

IDNA — Internationalizing Domain Names in Applications.

JSON — JavaScript Object Notation.

MCC — Matthews correlation coefficient.

MMD — Maximum mean discrepancy.

MPS — Metal Performance Shaders, the Apple transformer-execution backend.

NCSC — National Cyber Security Center (source designation in PhishVN).

OS — Operating system.

PCG64 — NumPy's 64-bit-output permuted congruential generator.

PSI — Population stability index.

RF — Random Forest.

RFC — Request for Comments; RFC 3986 specifies URI syntax.

ROC — Receiver operating characteristic.

RQ — Research question.

SHA-256 — Secure Hash Algorithm with a 256-bit digest.

SIGINT — Interrupt signal used in the software interruption checks.

TPR — True-positive rate.

UCI — University of California, Irvine; host of the Machine Learning Repository.

URI — Uniform Resource Identifier.

URL — Uniform resource locator.

UTF-8 — Unicode Transformation Format, 8-bit encoding.

Glossary of Terms

Admission sequence — Service-assigned order of admission. It is part of exact response agreement, not a predictive field.

Calibration — Agreement between stated predictive probabilities and observed reference-label frequencies; distinct from ranking and threshold performance.

Closed-loop timing — A workload in which each client starts another request after its previous request completes. It does not represent an unrestricted arrival process.

Clustered bootstrap — Resampling whole groups, such as registrable domains, rather than treating all records as independent.

Discrimination — Ability to rank positive reference-label cases above negative cases; distinct from calibration.

Frozen threshold — A decision boundary selected using development evidence and kept unchanged for the specified evaluation.

Invocation — An actual execution of a model, rather than a routing intention or an accepted HTTP request.

Metamorphic relation — A declared relation between transformed inputs and expected outputs, used to test a specific behavioral property.

Noninferiority margin — The prespecified maximum acceptable loss relative to a comparator for a defined outcome and uncertainty rule.

Paired run — Two comparator executions matched by experimental condition and pair identifier. Run-pair summaries, not individual requests, are resampled for the service-ratio interval.

Registrable domain — The public suffix plus the immediately preceding domain label, determined under the retained suffix rules and used as the grouping key.

Source shift — A change in collection source or its resulting population. It does not by itself establish chronological drift or changed prediction errors.

Tail latency — A high percentile of the measured request-latency distribution, such as p95 or p99; its success/error denominator must be specified.

Transport-neutral representation — The declared feature representation that removes sensitivity to the prescribed HTTP/HTTPS scheme-only transformation; it does not establish equivalence of live pages.

Chapter 1—Introduction

1.1 Background

CyberSentinel screens uniform resource locators (URLs) before a user or application follows them. An address may arrive through an application programming interface (API), a retrieval pipeline or a generated response. At that point, the gateway has the address but has not fetched or rendered its destination. It can inspect syntax, structure and character sequences. It cannot consult page content, and it makes no Domain Name System (DNS), WHOIS or reputation-service queries. This limited access makes lightweight screening possible while placing clear limits on what the detector can know.

CyberSentinel makes three design choices testable: what information to extract from a URL, when to invoke an expert and how to respond to changed inputs. Domain-separated testing evaluates unseen domains; evaluation across sources probes collection conventions that domain separation can preserve. Comparing both models on identical inputs establishes which decisions the expert can improve.

1.2 Problem Statement

An inline phishing gateway must detect threats, limit false warnings and respond promptly. A cheap first model saves computation; an expert may recognize patterns it misses. A distribution monitor flags inputs that differ from the training reference. CyberSentinel tests these components together under joint detection and service constraints. Each must earn its place through better decisions or lower cost.

The engineering question is which change addresses the observed behavior. Collection-specific cues, threshold transfer and shared model errors require different tests. Delay can also arise outside the classifiers. Paired decisions and controlled comparisons examine these possibilities separately. Chapter 3 defines the populations and reference-label limitations.

1.3 Purpose of the Study

The first stage of CyberSentinel is L1-regularized logistic regression on structural features. A compact character transformer replaces its decision for inputs within a validation-selected uncertainty band. A Gaussian mixture model (GMM) monitors those features and the first-stage score. During ordered offline replay, an alert changes routing for the next 256 requests. The evaluation follows those requests to determine whether the response to an alert helps detection. Separate HTTP measurements assess the service that carries out the decisions.

PhiUSIIL supplies model development and a domain-disjoint internal test; PhishVN v3.1.0, Mendeley repository Version 4, supplies external evaluation. Comparing models on identical URLs separates representation gains from changes in the evaluated population. Comparing both classes reveals the cost of those gains in false alerts. Following monitor activations into subsequent decisions tests whether escalation is useful. Chapters 4–5 connect these observations and the service measurements to research questions 1–3 (RQ1–RQ3).

Each research question leads to a design test. Paired model comparisons measure the feature gain; scheme interventions test format dependence. Monitoring replay follows

alerts into decisions, and routing comparisons count expert corrections. Shuffled-label controls examine validation ranking. Client and computation tests locate execution costs.

1.3.1 Thesis Statement

CyberSentinel demonstrates how to evaluate a URL gateway as a connected system: measure the information its features supply, the corrections its expert can make and the cost of execution. The experiments reveal a benchmark-format dependency, derive an exact routing constraint and verify a client-path improvement. These findings guide representation, expert and service design.

Section 1.4 states the research questions and numerical hypotheses. Chapter 4 reports their outcomes and uses controlled comparisons to explain them. Appendix C retains the working thesis and methodological record.

1.3.2 Research Objectives

The objectives are to measure what structural and character representations add across sources, whether monitoring improves decisions, and how selective expert use affects detection and execution. Fixed operating points support comparisons in both classes. Chapter 5 develops the resulting design priorities.

1.4 Research Questions and Hypotheses

The National Cyber Security Center (NCSC) source designation below is the publisher's label for a PhishVN positive stratum, not an independent adjudication performed by this study.

1.4.1 RQ1 and H1

RQ1: What incremental value do structural URL features and character-level representations provide under registrable-domain-disjoint and external evaluation?

H1: At the validation-selected FPR ceiling, full structural features improve recall over a length-only baseline, and the character-transformer cascade improves recall over Logistic-L1.

H1 is supported only if each of length-only, Logistic-L1, and cascade has observed FPR $\leq$ 1% on both PhiUSIIL group-test negatives and certified trusted-registry negatives, and if the registrable-domain-clustered 95% bootstrap lower bounds exceed zero for all four prespecified recall differences. The differences are Logistic-L1 minus length-only and cascade minus Logistic-L1, each evaluated on PhiUSIIL group-test positives and NCSC gold positives. A positive difference in only one source, or an improvement accompanied by excess FPR, does not support H1.

These are the only two primary H1 contrasts. Transformer-only remains a comparator and operational reference, not a third primary H1 gate. The cascade-minus-Logistic-L1 contrast measures the selective system contribution, not a pure causal isolation of representation.

1.4.2 RQ2 and H2

RQ2: Can GMM-based monitoring detect an external source/domain shift and guide escalation without exceeding the low-FPR operating constraint?

H2: GMM detects at least 80% of prespecified external shift windows with a false-alert rate of no more than 5%, and prospective routing reduces the false-negative rate relative to the fixed cascade while retaining FPR $\leq$ 1%.

H2 is supported only if all four conditions hold: external-window detection is at least 80%; the false-alert rate on the independent PhiUSIIL validation audit stream is no more than 5%; the alert policy's observed FPR on certified trusted-registry negatives is ≤ 1%; and the lower bound of the registrable-domain-clustered 95% bootstrap interval for false-negative-rate reduction on NCSC gold positives is greater than zero. The monitor measures departure in P(X). It does not by itself establish harmful drift, label shift, concept drift, or a causal change in model error.

1.4.3 RQ3 and H3

RQ3: What detection, escalation, throughput, and latency tradeoffs determine whether the fixed cascade is viable inline?

H3: For each system, observed FPR on certified trusted-registry negatives is ≤ 1%, and the Tranco reference-negative alert rate is ≤ 1% as a mandatory secondary safeguard. Cascade recall on NCSC gold positives is noninferior to transformer-only recall within 2 percentage points. On the 1%-prevalence benchmark reference, transformer invocation is at most 30%, real-HTTP p95 latency is at most 200 ms at concurrency 64, and the request-error rate is below 0.1%.

H3 requires both external-FPR safeguards for both the fixed cascade and transformer-only model. Exact one-sided 95% Clopper–Pearson upper confidence bounds accompany the two certified-registry FPR estimates and the two Tranco control alert-rate estimates. The domain-clustered 95% bootstrap lower bound for recall(cascade) minus recall(transformer-only) on NCSC gold positives must be at least −0.02. Stage 2 must be invoked for no more than 30% of the primary reference manifest. Pooled p95 latency

across the measured concurrency-64 runs must be ≤ 200 ms, and the request-error numerator divided by all 50,000 measured concurrency-64 requests must be < 0.1%.

Sections 1.4.1–1.4.3 define the original operating requirements and bootstrap specification. Section 3.8 supplies the paired uncertainty correction used in Tables 4.3 and 4.14 and the RQ/H answers. The correction preserves every metric and numerical margin; Appendix C retains the original estimates and decisions.

1.5 Engineering Contributions

For RQ1, structural features add 64.07 percentage points of internal recall. Validation labels also follow a URL-format convention, and changing the scheme changes frozen-model external decisions. A separate refit removes sensitivity across 8,622 pairs and measures the detection trade-off. These tests assess representation stability alongside accuracy, extending the robustness questions of Ahamed et al. (2026), Rashid et al. (2024) and Tsai et al. (2024). They do not assign a fraction of the gain to HTTPS (Sections 4.8 and 4.10.1).

For RQ2, the study follows monitoring alerts into decisions. The GMM recognizes 87.12% of external windows; escalation adds 218 certified false positives without another gold detection. Recognizing unfamiliar inputs and improving their classification require separate tests. Paired routing analysis counts available expert corrections, applying the downstream-error reasoning of Jitkrittum et al. (2023) to this gateway (Sections 4.8.1 and 4.10.2).

For RQ3, shared false positives set an exact 90.5487% floor on routing between unchanged decisions. Separately, worker-owned clients improve synthetic structural-only HTTP latency; a no-model control supports the client-path explanation. Comparable

predictions agree while admission order changes. Expert scoring cost is measured on its own workload. Building on established cascades (Li et al., 2021), these experiments evaluate both selective execution and service design.

1.6 Scope, Delimitations, and Limitations

Each retained URL is a detection unit, with an allow or alert response. The gateway does not block requests, remove content or take down destinations. Development partitions admit absolute HTTP and Hypertext Transfer Protocol Secure (HTTPS) URLs under the declared parsing rules. The terms raw-URL and URL-only refer to classification without fetching the destination; they do not imply that a publisher has preserved the original spelling. The amended external evaluation uses the publisher's exact url_norm (Section 3.4). Private request content, rendered pages, DNS, WHOIS, reputation services and network telemetry are outside the input boundary. Frontier artificial intelligence (AI) motivates this setting, but the experiments evaluate the URL service rather than a frontier model's reasoning, browsing or field deployment.

The study uses the publishers' source-based labels without independent adjudication. PhiUSIIL provides no public per-row source, timestamp, snapshot or independent-adjudication fields, limiting forensic and temporal interpretation. Preserved source files and explicit mapping, quarantine and source/domain-separation rules make the procedure reproducible; they cannot establish that the labels are correct. Under the disclosed amendment, the publisher's exact url_norm supplies external parsing and model input, while the original raw cells remain preserved. A scheme added by a publisher, or a record containing only a domain, cannot establish the transport or behavior of a live page.

The operating budgets give the study a definite standard for judging the gateway. They require FPR no greater than 1%, a two-percentage-point recall margin, expert invocation no greater than 30%, HTTP p95 no greater than 200 ms and request errors below 0.1%. Monitoring must reach 80% sensitivity with no more than 5% false alerts. A request times out after 2,000 ms. These requirements were fixed before confirmatory outcomes. Their values make the hypotheses testable, but they were not derived from a validated deployment cost model or a universal standard. Their suitability for a particular deployment therefore remains to be established.

Chapter 2 develops the literature-based rationale for the evaluation. Chapter 3 specifies the populations, estimands, model comparisons and mechanism tests. Chapter 4 connects their results, and Chapter 5 answers RQ1–RQ3 and draws the engineering implications. The appendices provide the evidence map, complete service schedule and methodological record.

Chapter 2—Review of the Literature

This focused, nonsystematic review explains CyberSentinel's design choices. URL detection determines what evidence the gateway uses; transfer research asks whether it survives a change of source. Cascades and monitoring inform when to spend computation and how to judge the resulting decisions. Service measurement examines the caller's experience.

The literature informs the design and interpretation of the study. Chapter 3 distinguishes the original hypothesis tests from diagnostic analyses, and the supporting record identifies when sources informed the investigation.

2.1 Raw-URL Phishing Detection

Phishing detectors can use page content, appearance, host infrastructure, reputation, email context or the URL itself (Basit et al., 2021; Khonji et al., 2013). At a pre-fetch application programming interface (API) boundary, however, the address may be the only available evidence. Its length, punctuation, digit ratios, entropy, Internet Protocol address literals, port syntax and authority forms can be inspected cheaply. The value of these indicators depends on the collection process and period represented in the data.

PhiUSIIL combines URLs from phishing sources with legitimate URLs drawn from Open PageRank (Prasad & Chandra, 2024a, 2024b). Its associated study uses webpage information in addition to URL features. CyberSentinel uses the URL and features calculated locally, so the two evaluations do not provide a like-for-like

performance comparison. The present evaluation retains the publisher's class assignments.

Detection is one part of the wider response to phishing. Khonji et al. (2013) consider it alongside prevention and corrective measures, while Basit et al. (2021) review machine-learning, deep-learning, hybrid and scenario-based approaches. An alert about a URL does not authenticate the page operator or establish the sender's intent. Its practical effect also depends on how the warning is handled. The present study addresses classification and service behavior within that wider problem.

Early URL research illustrates the tradeoff between local information and additional lookups. Ma et al. (2009) combine lexical and host properties to classify malicious addresses, a broader category than phishing. Sahingoz et al. (2019) compare classifiers using natural-language-processing features from URLs without third-party services. Both provide reasons to investigate a lightweight first stage. Its error rates on an external collection still require direct evaluation.

A browser or client-side detector may have access to evidence unavailable before retrieval. Jain and Gupta (2018) use URLs and page source without third-party dependencies; Rao and Pais (2019) also use source-code and third-party features. Marchal et al. (2017) evaluate Off-the-Hook as a browser add-on and study its warnings with users. Comparing their accuracy directly with CyberSentinel would mix differences in feature access, populations, labels and protocols. Retrieving further evidence and involving users may be valuable. The question addressed here is how much can be achieved from the URL before taking those additional steps.

2.2 Structural and Character Representations

Different models make different parts of an address available for learning. L1 regularization can suppress weak or redundant predictors while leaving a sparse model that can be inspected (Tibshirani, 1996). Random forests learn nonlinear feature combinations (Breiman, 2001). Character convolutional networks (Zhang et al., 2015), URLNet's character and word representations (Le et al., 2018), and attention-based models (Vaswani et al., 2017) instead offer ways to learn from sequences. Greater capacity does not by itself establish gateway value. The comparison must show whether that capacity produces useful changes in decisions.

Ahamed et al. (2026) evaluate URL models under adversarial, domain-disjoint and external conditions. On a separate PhishTank/Majestic collection, phishing recall remains high while false positives increase on the independent benign source. They also examine mutations in both classes and interpret the effects separately. CyberSentinel tests a more specific dependency by changing Hypertext Transfer Protocol (HTTP) and Hypertext Transfer Protocol Secure (HTTPS) prefixes while keeping structural weights fixed. A separate refit then removes scheme dependence across the affected feature pathways. Its evaluation checks both invariance and the resulting changes in true-positive detections and false positives.

The length-only model provides a simple starting point because it receives just one scalar. Logistic-L1 adds structural information, the transformer uses character sequences, and Random Forest supplies a secondary comparison for nonlinear structural combinations. The classical models use scikit-learn interfaces (Pedregosa et al., 2011). Chapter 3 fixes their capacity and training settings without an architecture search. The

findings therefore apply to the fitted models examined here. They do not identify the strongest possible transformer or establish the performance of URL detection generally.

2.3 Domain Leakage and Source Shift

Related records on both sides of a training–test split can make a model appear to generalize better than it does. Assigning each registrable domain to a single partition, without consulting labels, asks whether the model can work on domains outside its training allocation (Roberts et al., 2017). This addresses domain overlap while leaving possible source effects. Rashid et al. (2024) study degradation and adaptation across phishing datasets; Tsai et al. (2024) examine dataset bias and invariant representations for malicious URLs.

PhishVN's grouped splits and source/confidence tiers allow external results to be examined by provenance (Vu, 2026a). In this study, all retained records pass through the same label-blind routing procedure. The gold-positive and certified trusted-registry-negative strata are used afterward for primary analysis. Tranco supplies a popularity-based reference-negative control, whose ranking and potential manipulation affect its interpretation (Le Pochat et al., 2019). Popularity alone does not verify a negative label. Nor does a dataset's name establish the correctness of its labels.

The meaning of transfer depends on what changes between sources. Pan and Yang (2010) distinguish transfer settings, and Ben-David et al. (2010) relate target error to source error and distribution divergence under shared-hypothesis assumptions. Sugiyama et al. (2007) examine importance weighting when $P(X)$ changes while $P(Y|X)$ stays fixed. Collection and labeling practices can violate that condition. The present

evaluation measures frozen-model performance across declared source/domain strata; it applies neither model adaptation nor the Ben-David et al. bound.

2.4 Calibration at Low False-Positive Rates

When phishing is rare, false alarms can dominate the warnings users receive even if recall is high. Axelsson's (2000) base-rate analysis explains this concern, although it does not establish the study's 1% ceiling. Here, validation selects the highest recall satisfying a one-sided 95% Clopper–Pearson FPR upper bound ≤ 1%, with fixed tie rules and no test-informed relaxation (Clopper & Pearson, 1934). That binomial calculation requires an appropriate independence model. Section 3.8 addresses the uncertainty when URLs share domains.

Ranking, probability quality and operating-point error answer different questions. Receiver operating characteristic area under the curve (ROC AUC) measures pairwise ordering across thresholds; even narrowly separated scores can have high AUC. Precision –recall analysis is informative under class imbalance (Saito & Rehmsmeier, 2015), Matthews correlation coefficient (MCC) uses all four confusion categories (Chicco & Jurman, 2020), and proper scoring rules assess probabilities (Gneiting & Raftery, 2007). None substitutes for the false-positive and detection counts at the threshold a gateway will use. This distinction also matters when interpreting shuffled-label controls.

Prevalence changes the burden of alerts even when class-conditional error rates stay the same. The study's projections hold those measured rates fixed and vary the assumed class mixture. This resembles the label-shift setting considered by Lipton et al. (2018), but their correction is not applied and deployment prevalence is not estimated.

The projections show what follows from the assumed mixture. Ranking and probability calibration are examined separately.

2.5 Selective Cascades

A cascade lets a cheaper model handle some inputs so that an expensive model runs less often. CascadeBERT uses calibrated language-model cascades to trade computation against prediction quality (Li et al., 2021). Alajaji (2026) evaluates a validation-selected cascade for phishing email bodies, beginning with a classical model and referring selected inputs to a transformer. ExpertFusion provides related URL work (Hussain et al., 2027). Its repository describes fixed neural encoders, lexical features, prior-specific calibrated experts and fusion adapted using unlabeled target statistics (aI-area, 2026). Only that documented architecture is considered here; the full paper is not replicated and performance is not compared. These studies establish the precedent for calibrated deferral. CyberSentinel examines whether the selected expert calls improve URL decisions under a fixed FPR constraint and measures their cost.

The fixed cascade chooses its invocation rate from validation data. It considers symmetric bands at every unique absolute distance from the locked Logistic-L1 threshold and selects the band requiring the fewest expert calls. A feasible band must retain recall at least equal to transformer recall minus 0.02 and satisfy the FPR upper bound $\leq 1\%$. The test data play no role in choosing or revising that band.

Expert selection is a comparison of errors on the same input. Jitkrittum et al. (2023, Section 3.2) formalize a cost-sensitive decision using the difference between the models' conditional probabilities of correctness. Uncertainty in the first model need not identify an advantage for the second, particularly under specialization, label noise or

distribution shift. Ovadia et al. (2019) show that uncertainty estimates themselves can deteriorate under shift. The present study consequently measures available corrections, selected repairs and introduced errors at fixed thresholds, then measures expert cost separately. It evaluates this gateway's calls without claiming an optimal deferral rule.

2.6 Distribution Monitoring and Routing

A Gaussian mixture model (GMM) represents the monitor's reference distribution without using phishing labels in its fitting objective (Dempster et al., 1977). Diagonal covariance keeps the number of parameters limited. The Bayesian information criterion (BIC) balances fit and complexity to select one to six components (Schwarz, 1978). The monitor scores complete windows by their mean negative log-likelihood. Separate calibration and false-alert streams test whether the selected boundary holds beyond the windows used to choose it. The decisions following an alert then show whether escalation helps.

Detecting a change in inputs leaves the response to that change unresolved. CADE detects and explains drift samples in malware and intrusion settings (Yang et al., 2021), providing relevant security context without validating the likelihood model or routing rule used here. Gama et al. (2014) and Lu et al. (2019) likewise distinguish detection, interpretation and adaptation. A departure in P(X) alone says nothing conclusive about P(Y|X), labels or prediction errors. The gateway's response must therefore be evaluated through a separate comparison of errors.

An unfamiliar input can be benign, and a harmful input can resemble the fitted reference (Chandola et al., 2009). A distribution test such as maximum mean discrepancy (MMD) likewise cannot guarantee fewer phishing errors (Gretton et al., 2012). This study

reports window detection, reference alerts, request errors and additional model execution with their respective denominators. Overlapping windows and activations that affect future requests also restrict the use of inference based on independence.

2.7 Gap Addressed by the Study

CyberSentinel connects three engineering choices: URL representation, monitoring-guided expert selection and service implementation. Paired outcomes and controlled interventions measure how each choice affects the gateway. Table 2.1 places this contribution within robustness, adaptation and deferral research.

Table 2.1. Closest prior work and the boundary of the present contribution.

Closest work	**Setting and contribution**	**Specific comparison in this study**
Ahamed et al. (2026)	URL robustness, external specificity, and class-specific effects of transformations	Measures a frozen-weight scheme intervention, verifies all affected feature paths in a separate refit, and quantifies its detection tradeoff. External degradation and two-class testing are established prior work.
Rashid et al. (2024)	Cross-dataset phishing-URL generalization through unsupervised domain adaptation	Adapts across sources, whereas this study freezes the detector and prohibits external refitting or recalibration.
Tsai et al. (2024)	Diagnosis and adversarial mitigation of dataset bias in malicious-URL models	Studies invariant representation learning rather than a low-FPR cascade, monitor, or service evaluation.

Closest work	Setting and contribution	Specific comparison in this study
Li et al. (2021)	Calibrated complete-model cascades for general language-model inference	Establishes cascade prior art outside phishing and does not test the present URL or gateway constraints.
Alajaji (2026)	Validation-selected classical-first transformer deferral for phishing email bodies	Uses a different modality and does not establish raw-URL or measured inference-gateway behavior.
Jitkrittum et al. (2023)	Characterizes deferral using downstream error structure, including under shift	Measures development-selected URL repairs, external shared errors and selected harms at fixed operating points. The cohort-specific counting bound is not new deferral theory.
Hussain et al. (2027)	Calibrated URL-expert fusion and target-prior adaptation, as documented in the author repository (aI-area, 2026)	Measures selected expert repairs, introduced errors and actual execution cost for this cheap-first gateway. The architecture-level comparison does not establish full-paper coverage, priority or superior performance.
Yang et al. (2021)	Detection and explanation of individual security drift samples	Does not establish the present GMM, window construction, false-alert audit, or routing rule.

Structural-versus-length tests measure the feature gain; validation controls and scheme interventions examine its dependencies. Ordered replay follows alerts into decisions, while paired outputs count expert corrections and new errors. Separate scoring

and HTTP tests locate execution costs. Each comparison informs what to change and how to judge it.

2.8 Evaluation-Guided Engineering Iteration

Evaluation must account for what the analyst already knows. Arp et al. (2022) describe how spurious correlations, unsuitable evaluation sets and test-informed development can distort security machine-learning results. A diagnostic observation can justify a mechanism test without establishing the cause of the observation. TESSERACT also distinguishes spatial from temporal evaluation: excluding domains does not establish performance on future data (Pendlebury et al., 2019). Comparison D uses an additional 2020 benchmark that predates the reported PhiUSIIL phishing collection, making it retrospective. Shared upstream feeds further limit dataset independence.

Calibration and selection change how predictions are used, but need not improve their underlying ranking. Guo et al. (2017) calibrate confidence using validation data; in binary prediction, a monotone transformation with positive temperature preserves ROC ranking. SelectiveNet learns under a coverage/rejection constraint (Geifman & El-Yaniv, 2019). Neither algorithm is implemented here, and neither approach makes deferral sufficient to correct errors.

Metamorphic testing evaluates a specified relationship between an input and a transformed companion (Chen et al., 2018). In comparison D, each URL receives an opposite-scheme companion, and both must produce exactly equal features and scores. This directly checks the intended representation change. It does not establish that the corresponding live pages are equivalent or provide new labels. The labeled comparison

measures the detection consequences, while separate service measurements assess execution cost.

2.9 Operational Measurement and Evidence Units

A caller's wait includes connection management, scheduling, queueing and response validation as well as inference. Dean and Barroso (2013) explain the importance of the slow end of the latency distribution. Kalibera and Jones (2013) emphasize replication at the levels where variation occurs. Accordingly, service comparison S estimates uncertainty from paired run-level ratios and reports pooled request quantiles separately. More requests within a run increase its observations, but do not create independent runs.

Timing results depend on the workload and arrival process. A closed-loop client waits for a response before sending its next request; externally timed open arrivals can continue while earlier requests are unfinished (Schroeder et al., 2006). The measurements here describe the tested path without establishing unrestricted production capacity. Concurrent execution also requires care in comparing responses. Predictions may agree even when admission order changes. The study checks both, accounts for every attempt and error, and treats timed-out pairs as noncomparable. Detection, routing, computation and service results retain their own evidence units.

Chapter 3—Methodology

3.1 Research Design

Each research question is tested at the unit where its effect can be observed. RQ1 compares decisions on identical uniform resource locators (URLs), using the contrast-specific uncertainty in Section 3.8. RQ2 follows complete 256-request windows at stride 64 into subsequent decisions. RQ3 measures complete Hypertext Transfer Protocol (HTTP) requests across a 125-cell matrix. Models, thresholds and decision rules stay fixed across its two sessions; Section 4.6 and Appendix C describe the resulting limitations.

3.1.1 Architecture and Information Flow

Preparation determines which URLs are eligible and assigns their identities. Each URL supplies 25 structural features and a Logistic-L1 score. The uncertainty band decides whether to retain that decision or replace it with the character transformer's. The Gaussian mixture model (GMM) observes the features and first-stage score; its alerts affect only future requests. Publisher-label strata are joined to saved outcomes afterward for evaluation. HTTP execution is measured separately (Figure 3.1 and Table 3.1).

The router processes every mechanically eligible record in the same ordered stream. It never receives the external outcome strata: gold, certified and secondary designations enter only the later analysis. Saved records distinguish a request assigned to the transformer from an actual transformer execution, allowing routing decisions and computation to be measured separately.

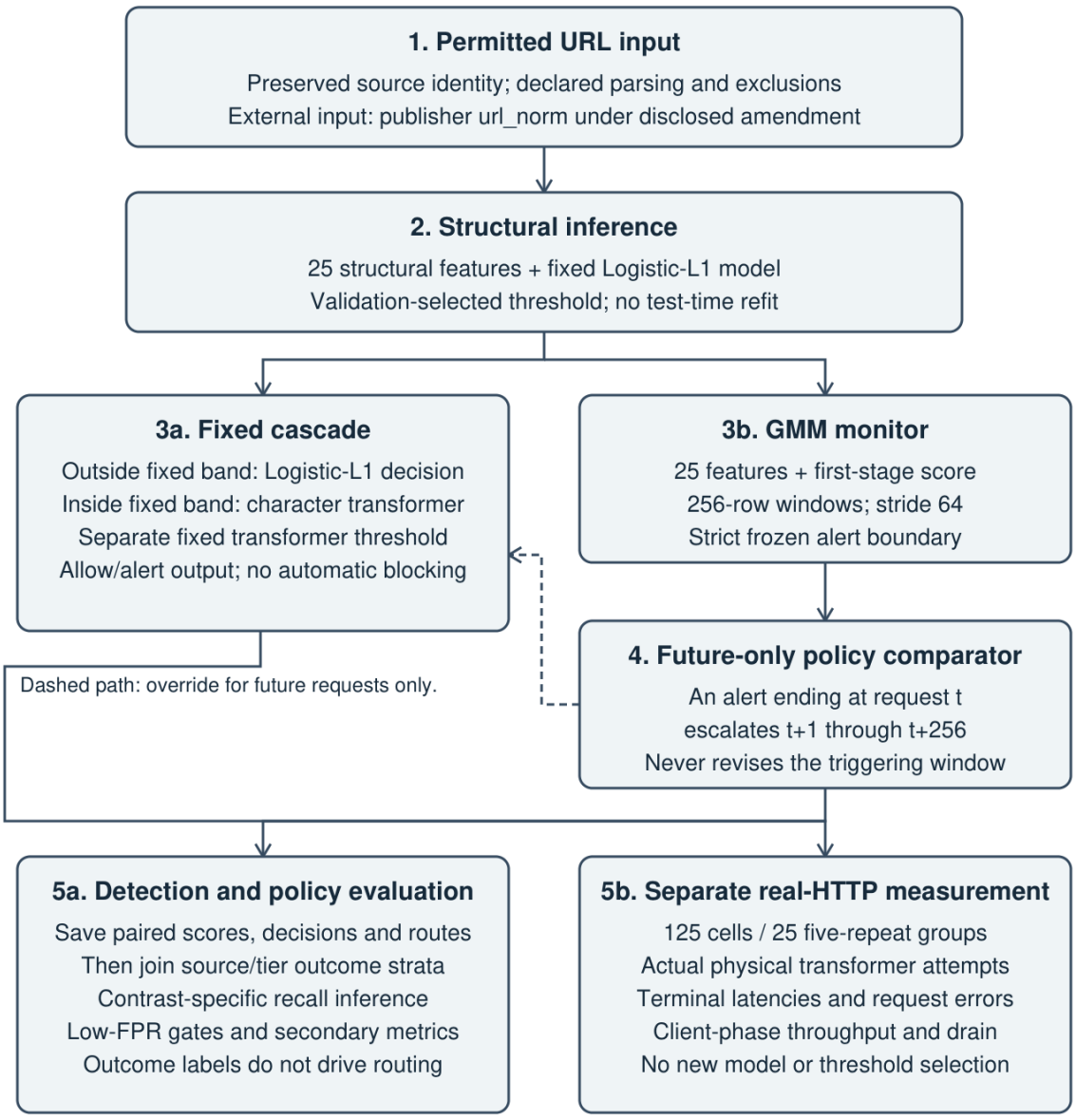


Figure 3.1. Frozen inference, future-only routing and separate evaluation boundaries.

Recall uncertainty uses the contrast-specific methods in Section 3.8.

Table 3.1. Reading map from technical component to experiment and evidence.

Plane	Technical operation	What its evidence establishes
Preparation	Preserve source fields; apply declared parsing, quarantine, domain allocation and overlap checks	Which records and domains enter each denominator; not independently adjudicated labels
Representation	Length-only, 25-feature Logistic-L1, character transformer and fixed cascade	Paired recall differences and specificity at unchanged thresholds; RQ1, Tables 4.2–4.3
Monitoring and policy	Training-fixed GMM; validation-calibrated windows; next-256 escalation	Distribution sensitivity, reference false alerts and the policy's separate error consequences; RQ2, Tables 4.4–4.5
Service	Real HTTP client/service processes; physical inference attempts; terminal outcomes and latencies	End-to-end cost, failures and throughput for every scheduled repeat; RQ3, Tables 4.6–4.7
Verification	Bound artifacts, retained-cell eligibility, process exits and saved-evidence recomputation	Traceability from accepted execution to all 22 decisions; not independent replication

3.2 Outcome-Label Contract

PhiUSIIL's label convention is the reverse of the local outcome field, so the mapping is explicit: native label 0 becomes is_phishing=1, and native label 1 becomes is_phishing=0. External categories are checked separately. National Cyber Security Center (NCSC) gold records provide the primary positives, and certified trusted registries provide the primary negatives. NCSC silver and ChongLuaDao/OpenPhish bronze

records are secondary strata. Tranco supplies a reference-negative control. These are publisher classifications without new forensic adjudication; Tranco is not verified benign ground truth.

External records are retained only when they have a valid URL, source, source class, confidence tier, published split and stable identifier. Unknown mappings are quarantined. An entire group is excluded if canonical labels conflict, its domains cross development or published external splits, or an external domain overlaps PhiUSIIL. For same-label duplicates, the record with the lexicographically smallest stable identifier is retained. Groups lacking such an identifier are quarantined. Eligibility cannot be changed through personal adjudication, and outcome strata are assigned after routing.

3.3 Development Source and Preservation

PhiUSIIL is the sole source for fitting, validation and internal evaluation. It is dataset 967 in the University of California, Irvine (UCI) Machine Learning Repository and is licensed under Creative Commons Attribution 4.0 (CC BY 4.0) (Prasad & Chandra, 2024b). Appendix C identifies the archive and comma-separated values (CSV) files. The publisher lists Open PageRank for legitimate URLs and PhishTank, OpenPhish and MalwareWorld for phishing URLs retrieved October 1, 2022–May 21, 2023 (Prasad & Chandra, 2024a). The legitimate-URL collection window has not been established. Independent per-row source, timestamp and adjudication fields are also unavailable, restricting source-specific and temporal conclusions.

3.4 Canonicalization and Quarantine

Preparation accepts absolute HTTP or Hypertext Transfer Protocol Secure (HTTPS) URLs. It lowercases the scheme, applies Internationalizing Domain Names in

Applications (IDNA) normalization to hostnames, normalizes numeric ports and removes default ports. Empty paths become '/', and percent-escape hex is uppercased without decoding. User information, path spelling, query order and fragments are preserved. The resulting canonical keys identify duplicates and conflicts; they do not supply labels. Unsupported inputs and conflicting groups receive recorded, mechanical exclusion reasons.

External parsing and model input use the publisher's exact url_norm field. Raw cells remain preserved, with unchanged mapping, overlap and quarantine rules and no local scheme repair, trimming or fallback. This choice followed initial preparation and label-count inspection but preceded the amended predictions (Section 4.6).

3.5 Registrable-Domain Allocation

Registrable domains are defined by the Public Suffix List snapshot in suffix-rules.dat. Each group is ranked using a Secure Hash Algorithm (SHA-256) digest of the allocation identifier, seed 20260816 and domain. Hamilton largest-remainder allocation targets 70% training, 15% validation and 15% group test, with ties resolved in that order. The allocation uses neither labels nor stratification and permits no rerolls. Training data fit the preprocessing and models; validation data select thresholds, checkpoints, bands and monitor boundaries. Group test is excluded from original fitting and selection. Its later exploratory reuse is disclosed in Section 4.8.1.

Internal evaluation establishes domain separation, not complete analyst blindness. Searches exposed group-test rows before scoring, and interruptions required repeated scoring under unchanged models and thresholds. Appendix C records these limitations; gate development treats the exposed outcomes as exploratory.

3.6 Detection Models

The extractor calculates 25 ordered float64 features from the supplied string, its urlsplit components and the IDNA-normalized hostname. It does not contact the destination. The features comprise six lengths, six hostname/path/query counts, eight composition/entropy/escaping quantities and five scheme/component indicators. Their names, order and parsing rules appear in the method supplement. Correlation among features prevents individual coefficients from being interpreted causally. A missing or invalid raw_url causes an error instead of imputation. Labels, split and record identities, canonical/domain keys and publisher provenance are excluded from the predictors.

Both length-only and Logistic-L1 use a training-fitted StandardScaler followed by LogisticRegression. The solver is SAGA with L1 regularization, C=1, balanced class weights, an unpenalized intercept, max_iter=5000, tol=1e-4 and random_state=42. Length-only receives raw codepoint length; Logistic-L1 receives all 25 features. A fit must converge before the iteration limit and have finite state. No hyperparameter search is conducted. Training-only diagnostics led to SAGA after liblinear did not satisfy the fitting checks, before accepted validation. The diagnostics, warnings and solver change are documented.

Scoring outputs must be finite. Three recorded numerical warnings are permitted only when decision values and the full probability matrix agree with independent float64 arithmetic to absolute and relative tolerances of 10^{-12}. Other warnings remain fatal. The authoritative scoring procedure preserves floating-point ties; Appendix C and the method supplement retain the warning and runtime specifications.

Stage 2 first canonicalizes the URL. It encodes non-American Standard Code for Information Interchange (ASCII) characters as Unicode Transformation Format, 8-bit (UTF-8) bytes, then percent-encodes those bytes while preserving existing escapes and Request for Comments (RFC) 3986 reserved characters (Berners-Lee et al., 2005). A sequence longer than 256 characters retains its first 192 and last 64. The training-only ASCII vocabulary reserves padding (PAD)=0 and unknown-character (UNK)=1, with right-padding masked. The float32 PyTorch 2.7.1 transformer has 192-wide token/position embeddings, four pre-normalization layers, six heads, feed-forward width 768, Gaussian error linear unit (GELU) activation and dropout 0.1. Final LayerNorm and masked mean pooling supply a linear logit head.

Deterministic training on Apple's Metal Performance Shaders (MPS) backend uses seed 42 and class-weighted binary cross-entropy. AdamW uses learning rate 10^{-4} and weight decay 0.01, with no scheduler (Loshchilov & Hutter, 2019). Training and validation batches contain 256 and 512 examples, respectively; gradient clipping is 1.0 and training is capped at 40 epochs. The supplement specifies initialization, optimizer constants and data-loader behavior.

Validation average precision (AP) is evaluated after each epoch with dropout disabled. Improvement must exceed 10^{-4}; five consecutive nonqualifying epochs stop training, and the earliest qualifying best checkpoint is restored. Nonfinite values, missing classes or deterministic-operation errors halt publication. Stage 1 remains the accepted Logistic-L1 artifact. The supplement retains the complete acceptance and serialization rules.

3.7 Threshold and Cascade Calibration

A score at or above the selected threshold triggers an alert. Scores are not assumed to be calibrated estimates of deployment risk. Threshold candidates comprise the unique validation scores and nextafter(max_score,+infinity). Selection maximizes recall while requiring the one-sided exact 95% Clopper–Pearson false-positive-rate (FPR) upper bound to be ≤ 1%. Ties favor the smaller bound and then the higher threshold. If no candidate meets the constraint, the result is recorded as target_not_met; the constraint is not relaxed.

Cascade calibration follows selection of the component thresholds. It considers inclusive symmetric bands satisfying abs(stage1_score-stage1_threshold) ≤ half_width, with widths drawn from the unique validation distances. Within the band, the transformer replaces stage 1's score and decision. Outside it, Logistic-L1 supplies both. A feasible band must achieve recall ≥ transformer recall minus 0.02 and meet the same FPR-bound constraint. Selection minimizes calls and then width. The thresholds and selected band remain fixed throughout evaluation and replay.

3.8 RQ1 Analysis

OpenBLAS/MPS scoring processes each record individually. A development comparison identified one transformer decision that changed with batch scoring, motivating this singleton convention. Weights, thresholds and the routing band were retained without recalibration; all accepted paired predictions use singleton scoring. Equivalence to the earlier batch procedure and its checkpoint selection has not been established. Appendix C records the scoring amendment.

Every system is compared on the same aligned records with models and thresholds held fixed. Section 4.6 and Appendix C document the recovery procedures and prior information exposure that affect interpretation.

Bootstrap resampling follows the general framework of Efron (1979), with the domain-clustered estimand specified below. The original primary recall analysis used a registrable-domain-clustered percentile bootstrap using 2,000 replicates and seed 20260816. H1 uses the lower confidence bounds for the four prespecified differences and the observed FPR gates in Section 1.4.1. Average precision (AP), precision, F_2 score, Matthews correlation coefficient (MCC), balanced accuracy, receiver operating characteristic area under the curve (ROC AUC), Brier score, and calibration error are secondary descriptions.

The paired recall calculation starts from saved binary predictions for the specified positive stratum. Identifiers must be unique and aligned across prediction vectors. Let D denote the number of registrable domains, ordered by canonical ASCII spelling. Each replicate draws D domains uniformly with replacement and includes every eligible positive row for each sampled occurrence. The sum of candidate-minus-reference decisions is divided by the sampled row count. This retains URL weighting when domains differ in size. Averaging domain-level recalls would estimate a different quantity.

Every contrast uses 2,000 replicates with seed 20260816. Contrasts within the same stratum share the same domain draws. The interval endpoints are the linearly interpolated 2.5th and 97.5th percentiles. Empty strata have no estimate, and a single domain cannot support a clustered interval. The supplement specifies the generator and

implementation. The correction that follows explains how degenerate bootstrap results are interpreted.

When every observed prediction pair agrees, resampling can only reproduce the same difference. It cannot reveal an error pattern that did not occur in the sample. A zero-width bootstrap interval therefore does not establish a two-percentage-point population recall margin. The paired method retains the same research questions, populations, numerical margins and two-sided 95% convention. The replacement method was introduced after the study and was not prespecified.

For a nonstateful gold comparison, b counts candidate-only detections and c counts reference-only detections among n positive pairs. Their difference, (b-c)/n, estimates the difference in detection probabilities. With independent, identically distributed pairs, each discordant count has a binomial marginal. Clopper–Pearson bounds allocate 0.05/4 to each of the four marginal tails. Projecting their simultaneous rectangle gives [L(b)-U(c), U(b)-L(c)]. The union bound provides at least 95% coverage without requiring b and c to be independent. This conservative envelope is not the shortest exact interval. Newcombe's (1998) method 10 supplies an approximate-score sensitivity. For 69 concordant pairs, the envelope is [−6.1533, 6.1533] percentage points and method 10 gives [−5.2737, 5.2737]. Neither establishes the −2-point requirement, and neither demonstrates inferiority.

The uncertainty methods apply to different dependence structures. The 69 gold rows belong to distinct domains, but distinct domains establish neither independent campaigns nor accurate labels. Both are assumptions of the gold-pair intervals. The internal structural-versus-length comparison instead uses whole-domain bootstrapping

with URL weighting because domains repeat. Independent-row formulas cannot replace that calculation or support inference for a stateful policy. Although the internal cascade-minus-Logistic-L1 and gold policy-minus-cascade estimates are zero, neither has an established population lower bound. Historical row-wise FPR bounds have a related limitation: the 2,497 certified negatives occupy 234 domains. Those bounds therefore offer no population-coverage guarantee, and decisions use observed rates. Intervals apply to their individual contrasts rather than jointly across the 22 checks. Every condition of a joint hypothesis must still hold.

3.9 GMM Monitoring and Future-Only Routing

The monitor uses the 25 structural features and the accepted Logistic-L1 probability. Standardization is fitted using training-only population variance, with constant columns assigned scale one. The probability calculation continues to use the accepted model without refitting. Validation domains are hash-ranked with seed 20260816 and divided into disjoint calibration and false-alert audit streams. Source order is preserved within each. The supplement specifies the numerical backend and single-thread runtime.

Six diagonal GMMs are fitted on training data, using one to six components and five k-means starts apiece. The model with the lowest training Bayesian information criterion (BIC) is selected; ties favor fewer components (Schwarz, 1978). All candidates must converge with finite, valid parameters. A failed candidate cannot be skipped or retried. The supplement specifies the seed, tolerances, covariance regularization, iteration limit and numerical-error policy.

The monitor calculates mean negative log-likelihood over complete 256-request windows at stride 64. A window triggers an alert only when its score strictly exceeds the linearly interpolated 95th percentile of the calibration scores. The audit passes if 20 × alerts ≤ complete windows, without retuning. Every complete external window is a prespecified shift observation, including overlapping windows. Incomplete terminal windows are excluded. Neither rate is assigned an independent-binomial interval.

External requests are replayed in file order. The first 256 use the fixed cascade; subsequent alerts affect only the next 256 requests. Overlapping activations are combined, truncated at the stream's end and limited to one transformer call per request. For the saved outcomes, policy-minus-cascade recall equals the reduction in false-negative rate. Resampling these outcomes does not rerun the router. The comparison is offline, nonrandomized and conditional on the observed order, with dependence extending beyond shared domains. A change in P(X) need not represent a harmful change in P(Y|X). Maximum mean discrepancy (MMD), population stability index (PSI) and string probes are secondary analyses.

3.10 External Evaluation Freeze

The complete retained PhishVN test split is routed in file order before outcome strata are applied. No model is refitted or recalibrated for this evaluation (Vu, 2026b). PhishVN v3.1.0 is distributed as Mendeley Version 4. The metadata describe a documentation-only update from Version 3, retaining byte-identical data and adding label-audit documentation.

The initial freeze specifies sources, populations, artifacts, thresholds, gate, monitor, evaluator, manifest and runtime. Input-field amendment followed preparation

and label-count inspection; recovery followed predictions and partial measurement. Only the initial freeze was prediction-blind. Section 4.6 and Appendix C document these limitations and the unchanged scientific rules.

3.11 RQ2 Analysis

RQ2 evaluates the response as well as the alert. Alongside external-window recognition, Section 1.4.2 requires a validation-audit false-alert limit, certified-negative policy FPR and positive paired recall improvement. The original 2,000-replicate domain bootstrap describes saved outcomes conditional on stream order; Section 3.8 corrects the uncertainty claims it cannot support. Following each alert into subsequent decisions measures the response without assigning its causal effect solely to the monitor signal.

3.12 RQ3 Operational Replay

The primary manifest contains 10,000 group-test URLs sampled without replacement at a designed 1% positive prevalence. The 0.1% and 5% mixtures are sensitivity workloads, not production-prevalence estimates. Separate 64-bit-output permuted congruential generator (PCG64) streams with seed 20260816 determine class sampling and order. Candidates are sorted by stable source-row ID before sampling. Every repeat uses the resulting manifest order. The first 1,000 rows are replayed as excluded warmup, followed by the entire measured manifest. An insufficient eligible yield permits neither sampling with replacement nor a changed target. Manifest identities bind the raw spellings.

The operational matrix contains 125 cells arranged in 25 groups of five repeats. It includes 90 fixed-cascade cells across three prevalences and concurrency 1, 8, 16, 32, 64 and 128, plus 30 transformer-only cells at 1% and five serialized live-shift cells. Ordinary

arms contain 1,000 warmups and 10,000 measured requests. Live-shift arms measure 8,701 external rows after a separate warmup and reset. H3 uses all 50,000 measured fixed-cascade requests at 1% prevalence and concurrency 64. Its pooled p95 includes failures and is calculated as a linear quantile, rather than an average of run quantiles.

Each run starts a fresh service and client, with one inference owner and a first-in, first-out (FIFO) queue holding up to 128 waiting requests. Closed-loop workers wait for a request to terminate before sending the next (Schroeder et al., 2006). Timing covers the connection, queue, processing, body reading and response validation; unsent backlog is excluded. Success requires HTTP 200 and valid, identity-matched JavaScript Object Notation (JSON) within 2,000 ms. There are no retries or clamped terminal latencies. Timeouts, malformed responses, transport failures and other statuses count as errors in the full denominator. Admitted work continues after a timeout. Phase closure stops late admissions and drains the remaining work before counters are recorded.

Physical transformer invocation is measured in the designated first concurrency-1 primary run. H3 applies the limits in Section 1.4.3. All primary runs belong to the first session; one descriptive group crosses the two-session boundary, confounding session with workload order. Section 4.6 and Appendix C retain the cell allocation and execution record.

3.13 Reproducibility and Integrity Controls

The record lets a reader trace a result from input to outcome. Appendix A links each finding to its population, fitted model and evaluation procedure. Separate calculations check saved counts, paired decisions and latency. Every declared diagnostic

fit and its outcomes are retained. Appendix C records information exposure and execution history so others can interpret and reuse the comparisons.

Secondary analyses examine formatting-only and Random Forest comparators, five historical shuffled-label fits, seed/runtime sensitivity, MMD/PSI, label-free probes, calibration and paired tests. They report AP, ROC AUC, uninterpolated recall at FPR $\leq$ 1%, Brier score and ten-bin expected calibration error (ECE). Scores and operating-point decisions remain distinct because the cascade stages use different thresholds. Four exact McNemar contrasts compare structural with length-only and cascade with structural in both positive populations. All Holm-adjustment slots are retained (McNemar, 1947; Holm, 1979). These nominal tests do not model domain dependence and do not replace the primary intervals.

A source-only classifier would require per-row provenance independent of outcome. PhiUSIIL lacks that information, and deriving source from class would make the comparison circular. The formatting comparator instead uses five indicators. Historical label permutations use fresh PCG64 seeds 42–46, fixed solver seed 42, unchanged validation labels and separate Clopper–Pearson (CP) cutoffs. Random Forest uses its accepted direct-leaf arithmetic. Training-fitted MMD/PSI references keep the monitor's separate calibration and audit streams, with strict 95th-percentile boundaries of their own. These comparators do not replace the primary artifacts.

The transformer sensitivity adds seeds 43–46 using the training-only vocabulary, original 256/512 batching and stopping rule. Seed 42 is not refitted. Every checkpoint and validation vector is retained, and all five seeds receive singleton secondary scoring with individually selected thresholds and bands. Because seed 42 was trained under an

earlier runtime, the differences cannot be attributed solely to seed. Probe streams retain every position, including no-ops and ineligible strings, without inheriting correctness labels. Models, boundaries and scalers remain fixed, and the monitor history resets for each stream. Section 3.16 audits the inputs consumed by the new shuffled-label fits; that verification does not extend to the historical fits.

3.14 Ethical and Safety Considerations

URLs are inert strings; no destination is contacted. Licensed original records are separated from private execution files. The domain-balanced routing cohort's URLs, domains, identifiers and predictions remain private; public aggregates, code and specifications support review. The gateway alerts for review without blocking requests or removing destinations.

Manual review is confined to a separately reported, post hoc description of errors. It cannot assign or override labels, change eligibility or quarantine, or alter thresholds, features, models, gates or hypothesis decisions.

3.15 Representation and Service Evaluation

3.15.1 Comparison Objectives and Scope

The initial evaluation motivated two controlled comparisons: D tests scheme invariance and detection; S tests client ownership, latency and response agreement. Specifications were fixed before candidate fitting, additional-benchmark predictions and primary client measurements. Each comparison uses a two-sided 97.5% bootstrap interval, dividing the error allowance without requiring independence; finite-sample coverage remains approximate. The transformer, GMM and H1–H3 criteria remain unchanged.

3.15.2 Additional Benchmark and Eligibility

The additional source is the CC BY 4.0 Web page phishing detection dataset, Mendeley Version 3. It contains 11,430 URLs split equally between the publisher-labeled classes (Hannousse & Yahiouche, 2020, 2021a, 2021b). Only URL and class are used. The 87 features, stored pages and live destinations are excluded. The metadata and inspected preprint give conflicting collection months, so the source is described as the 2020 benchmark. Its Alexa/Yandex legitimate-source mechanisms differ partly from development, while PhishTank/OpenPhish feeds overlap. It provides retrospective reference-label evidence rather than an independent production sample.

Eligibility is determined before scoring with the original canonicalization and suffix rules. Invalid URLs, mixed-label canonical duplicates and domains already present in the full retained PhiUSIIL/PhishVN manifests are excluded. Parseable domains from both raw and normalized PhishVN fields contribute to screening. The 50,832 unparseable field instances must not be interpreted as a count of missing records. Same-label duplicates retain first source order. Admission requires at least 1,000 rows and 250 domains in each class, although those minima do not guarantee statistical power.

Evaluation includes all 8,622 eligible rows: 4,651 positives from 2,897 domains and 3,971 negatives from 3,377 domains. One domain occurs in both classes, giving 6,273 domains overall. Figure 3.2 distinguishes development, diagnosis, screening and evaluation. The original training and validation partitions fit and calibrate the candidate. The additional benchmark selects neither the candidate nor its operating point, and no sampling depends on outcomes.

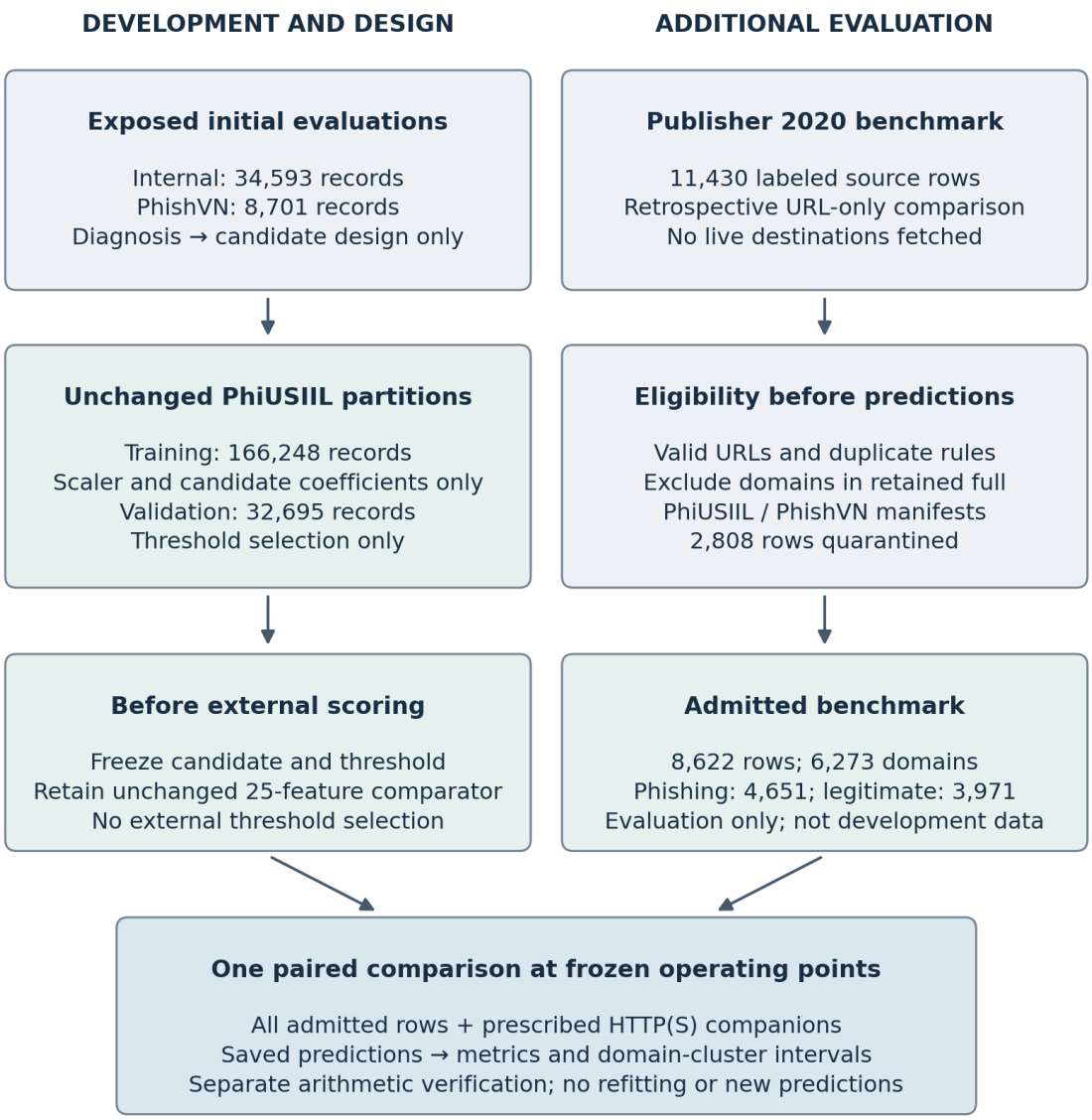


Figure 3.2. Source and partition roles in the bounded detection comparison. Arrows show permitted information flow; domain exclusion does not establish temporal or source-mechanism independence.

3.15.3 Transport-Neutral Structural Representation

The accepted 25-feature comparator is retained without refitting at threshold 0.2670846328466124. The candidate first validates an absolute HTTP/HTTPS URL, replaces only its scheme spelling with http, and calculates the original features apart from

is_https. The resulting 24-feature representation removes scheme effects from lengths, counts, ratios, entropy and the indicator. The original raw evidence remains preserved.

The candidate is fitted once on the original 166,248 training rows using training-only scaling and the Logistic-L1 settings in Section 3.6, with one thread and seed 42. Only the original 32,695 validation rows enter the unchanged maximum-recall/FPR-bound selection rule. Fitting converges in 4,971 iterations. At threshold 0.6815549262616749, validation gives TP=7,588/12,486 and FP=178/20,209, corresponding to 60.77% recall, 0.8808% FPR and a 0.9968% upper bound. These results determine the operating point; they do not establish external efficacy.

Every admitted URL receives a companion with the opposite scheme. The conformance check requires exactly equal features, scores and decisions under both singleton sklearn scoring and an independent float64 audit. It tests the representation, without supplying new website labels or establishing live-page equivalence. Because D combines normalization, refitting and recalibration, the comparison cannot isolate the causal effect of any one feature.

3.15.4 Detection Estimands and Uncertainty

D requires a recall gain ≥ 5 percentage points, a positive lower endpoint for its two-sided 97.5% interval, observed FPR $\leq 1\%$ and exact scheme invariance. Uncertainty is calculated from 10,000 paired domain resamples of the original eligible strings. A 64-bit-output permuted congruential generator (PCG64) with seed 20261001 samples the 6,273 sorted domains, including all rows each time a domain is drawn. Recall and FPR retain row weighting. Resamples with undefined class denominators are counted and excluded only from the affected intervals. Confusion matrices, precision, ROC AUC, AP,

Brier score and ten equal-width calibration bins are secondary measures. Undefined values are left undefined.

3.15.5 Service Comparison and Its Boundary

S compares a shared persistent client pool with a persistent client owned by each closed-loop worker. It holds the endpoint, input order, response contract, two-second deadline and no-retry rule fixed. The no-model and singleton structural workloads each contain 10,000 synthetic .example URL occurrences. They carry no labels or asserted prevalence, and neither calls the transformer. The no-model control helps distinguish classifier work from the wider client/service cost, although it cannot isolate individual scheduling costs.

The schedule crosses concurrency levels 1 and 64 with ten pairs for each workload/concurrency combination and two client topologies, producing 80 fresh-service arms. Odd and even pairs reverse the topology order. Each arm includes 1,000 warmups, 10,000 measured attempts and phase drains. The interrupted predecessor contributes no primary estimate. Its replacement rule was established before primary structural concurrency-64 testing (Appendix C).

The primary S statistic is the median worker/shared ratio of run p95 across ten structural concurrency-64 pairs. Ten thousand paired PCG64 resamples, using seed 20261002, produce a two-sided 97.5% interval. Only ten pairs are available, limiting bootstrap resolution. S requires a ratio ≤ 0.8, an upper endpoint < 1, pooled successful worker p95 ≤ 200 ms, errors $< 0.1\%$ across 100,000 worker requests, and exact paired responses after excluding request IDs alone. The last condition includes admission sequence. Agreement restricted to prediction fields is reported separately.

Retained observations account for successes, errors, latency, throughput, actual computation and phase drains. Host checks are sampled rather than continuous. S measures synthetic structural-only closed-loop HTTP behavior, with no claim of transformer acceleration or production capacity.

3.16 Diagnostic and Routing Comparisons

The initial results motivated four diagnostic comparisons with specifications fixed before execution. F tests scheme spelling under fixed weights; R tests expert selection on a separate domain-balanced cohort; C measures detector computation; P examines shuffled-label ranking. Each retains its own population and decision rule, with accepted models and H1–H3 and D/S requirements unchanged. Section 3.16.3 specifies the additional control checks.

3.16.1 Frozen-model Intervention and Domain-balanced Sampling

F uses the 8,701 previously evaluated external strings, retaining their labels and the structural weights, scaler and threshold 0.2670846328466124. Each original spelling is compared with versions forced to http: and https:, preserving every character after the first colon. Each version is scored once, and the original must reproduce its saved result. The paired features, scores and decisions show how spelling affects the frozen model. They do not describe live websites or isolate one feature, because the prefix affects several predictors.

R draws from retained PhishVN training and validation records, excluding publisher test (Vu, 2026a, 2026b). Both classes must have gold tier and a nonblank label_source. Positives require tinnhiemmang with a phishing label; negatives require tinnhiem_org or tinnhiem_web with a benign label. Invalid URLs and contradictory

canonical labels are rejected. After lowercasing and removal of trailing periods, normalized hostnames must match the publisher domains. Screening excludes publisher test, cross-split domains, PhiUSIIL, D and all discovered historical exposure. This local domain separation cannot establish independent campaigns, temporal novelty or absence of exposure elsewhere.

Every eligible domain contributes one URL: 1,392 positive domains and 1,262 negative domains. The smallest fixed-prefix SHA-256 rank selects the URL, with ties broken deterministically, and a separate domain hash sets the order. This domain-balanced retrospective sample changes the estimand from that of the original row-weighted study. The model receives exact publisher url_norm; canonical identities are used only for eligibility and overlap checks. Predictions play no role in inclusion, no domains are added after scoring, and labels select no model, threshold or gate.

3.16.2 Routing Comparisons and Paired Inference

R scores the accepted structural and transformer models once per record with frozen singleton conventions and thresholds. The transformer threshold is 0.033976949751377106. Saved decisions define structural-only, always-transformer, the original zero-width cascade, a symmetric half-width 0.20 gate and a directional 0.20 gate below the structural threshold. The bounded replay in Section 4.8.1 selected the primary symmetric and secondary directional rules before R predictions. Identity-hash controls match each revised rule's call count without using correctness information. These are finite-batch comparisons rather than online routers.

R tests whether original- and symmetric-cascade recall are noninferior to transformer-only under the two-percentage-point margin. It also tests whether the

symmetric rule improves recall over structural-only while retaining observed FPR ≤ 1% and calls ≤ 30%. Each contrast uses the conservative two-sided 95% discordance envelope in Section 3.8. Newcombe's (1998) method 10 is a prespecified sensitivity. The individual conditional intervals do not provide simultaneous coverage. Counts, logical calls, repairs and harms in both classes, and ranking measures are retained for every system.

The sample contains the fixed eligible yield; records were not added after scoring to obtain a passing result. Achieving the 90% power target depends on disagreement between the compared systems. With 1,392 independent positive pairs, planning power is approximately 98.25% under the exposed internal symmetric-rule discordance proportions. It falls to 53.76% when recall is equal and total discordance is 5%. Inference therefore uses the observed outcomes and stated assumptions. A descriptive sensitivity analysis resamples positive source-event-date groups through 2,000 fixed PCG64 replicates. These dates are not verified campaigns, and a degenerate bootstrap cannot replace the exact envelope.

3.16.3 Actual Detector Work and Audited Controls

Before inspecting scores, C selects 10 positive and 990 negative R strings in cohort-identity order. The resulting 1% mixture is a workload reference, not an estimate of traffic prevalence. Six paths run five times: structural, transformer, original cascade, symmetric, directional and the symmetric matched-count control. Each arm contains 100 excluded warmups and 1,000 measured requests at concurrency one. Five cyclic rotations do not fully balance positions. Timing covers input processing, gate work and actual expert scoring, but excludes model loading, evidence writes and admission checks.

Warmup and measured counts are kept separate. This experiment measures computation rather than HTTP performance.

For each arm, C records p50/p95/p99, summed scoring time, scores, decisions and actual stage starts and completions. Each repetition's summed time is divided by the structural-path total. The five paired ratios are summarized by their median and range, without a run-level confidence interval or production-capacity estimate. Outputs must reproduce R's saved component scores. Always-transformer provides a reference for expert computation cost; it does not represent selectively routed traffic.

P fits Logistic-L1 once for each seed 42–46, retaining training identities and shuffling only the ordered labels. The estimator.fit receipt identifies the consumed labels, row order, class counts, permutation and transformed input. Independent checks verify reloaded state and saved validation predictions. Validation remains unchanged at 32,695 rows, comprising 12,486 positives and 20,209 negatives. True validation labels select descriptive operating points under the existing Clopper–Pearson constraint. Those points therefore cannot support independent test-efficacy claims. The five fits neither constitute a formal permutation significance test (Ojala & Garriga, 2010) nor select a favorable seed or replace historical fits.

P tests three explanations for the ranking spread: an input or scoring error, incomplete numerical optimization, and a small random fitted direction aligned with structured validation labels. The audit checks partition hashes, domain/identity/canonical separation, actual consumed label permutations, training-only scaling and saved-score reconstruction, then independently recalculates tie-aware AUC. Subtracting the explicit is_https logit contribution from each frozen model measures sensitivity to that term

without refitting or changing another coefficient. It is not a causal importance estimate because correlated scheme information can remain in other predictors.

The ranking pattern in the five audited P fits motivated the additional permutation and convergence checks. Their fixed conditional randomization protocol schedules 99 additional training-label permutations, PCG64 seeds 1000–1098. It keeps the 166,248 training rows, 32,695 validation rows, training-fitted scaler, estimator settings and solver seed fixed. Every attempt is retained, including warnings or failures, with no seed substitutions or outcome-driven extension. True validation labels remain fixed and no threshold is selected. A reference comparison independently permutes validation labels using seed 210000 plus the fit seed. Constant-score AUC, removal of the HTTPS term and inverse-domain-size weighting provide further checks. Empirical quantiles summarize this fixed-data algorithmic distribution; they are not confidence intervals for new populations or a test certifying the absence of leakage (Ojala & Garriga, 2010).

A separate frozen convergence check continues each of the five saved fits from its coefficients and intercept, preserving the permuted labels and objective. Only tolerance and the iteration budget change: tol=1e-7 and max_iter=20000, with warm_start=True. Each seed is attempted once. The diagnostic reports AUC changes and first-order optimality residuals for the sum of class-weighted logistic losses and an L1 penalty weighted by 1/C, with an unpenalized intercept. Dividing residuals by total sample weight puts them on a mean-loss scale. This tests sensitivity to the original stopping tolerance; it does not replace the original results.

Chapter 4—Results

The evaluation measures CyberSentinel's component contributions and investigates their mechanisms. Sections 4.1–4.7 answer RQ1–RQ3 through representation, monitoring and service tests and adjudicate H1–H3. Sections 4.8–4.10 examine scheme sensitivity, expert corrections, client costs and shuffled-label ranking, connecting the findings to design decisions.

4.1 Completed Evaluation and Populations

The completed evaluation includes 34,593 internal records, 8,701 external records and 125 operational cells arranged in 25 groups of five repeats. Section 4.7 reports all 22 criterion decisions using the corrected paired uncertainty.

Saved confusion counts, paired outcomes and request records allow the arithmetic to be checked independently. Section 4.6 describes the two-session operational matrix; Appendix C retains execution identities and the verification record.

The corrected paired uncertainty follows Section 3.8; the observed predictions, populations and numerical margins are unchanged.

PhiUSIIL preparation retained 233,536 of 235,795 rows from 197,105 registrable domains. Exclusions comprised 1,380 invalid or unsupported uniform resource locators (URLs), 877 same-label canonical duplicates and two rows in one conflicting canonical group. The training partition contains 166,248 rows from 137,973 domains. Validation contains 32,695 rows from 29,566 domains, and group test contains 34,593 rows from 29,566 domains. Group test has 14,326 positives, 20,267 negatives and 9,757 domains in

its positive stratum. The outcomes retain the publisher's reference labels without new adjudication.

Amended preparation excluded 240 of the 8,941 published external test records, retaining 8,701 from 4,150 domains. The primary strata comprise 69 National Cyber Security Center (NCSC) gold positives from 69 domains and 2,497 certified-registry negatives from 234 domains. Secondary strata contain 417 NCSC silver positives from 412 domains and 4,555 Chongluadao/OpenPhish bronze positives from 2,294 domains. There are also 1,163 label-free Tranco controls from 1,163 domains. Because domains can occur across strata, their counts need not sum to 4,150. Routing processes every retained record before stratification. Table 4.1 describes this source composition; it does not estimate deployment prevalence.

Table 4.1. Retained external source, tier and outcome composition.

Source / tier	Role	Rows	Domains	Reference outcome
chongluadao openphish / bronze	secondary	4555	2294	phishing (1)
ncsc / gold	gold	69	69	phishing (1)
ncsc / silver	secondary	417	412	phishing (1)
tranco / control	tranco	1163	1163	unlabeled control
trusted registry / certified	certified	2497	234	legitimate (0)

Preparation examined 53,116 publisher rows, including 38 invalid-URL and 1,423 PhiUSIIL-overlap exclusions across all splits. Test-only exclusions are recorded

separately. Section 4.6 explains the use of exact publisher url_norm; no local repair or label reinterpretation is applied.

4.2 RQ1: Representation Value and Generalization

Table 4.2 reports fixed operating points, including the Gaussian mixture model (GMM) routing policy. Table 4.2a gives the confusion counts, and Table 4.2b gives the corresponding rates. True positives (TP), false positives (FP), true negatives (TN) and false negatives (FN) denote counts against the source reference labels. Internal recall uses P=14,326 and false-positive rate (FPR) uses N=20,267. External recall uses the 69 gold positives; external FPR uses the 2,497 certified negatives. TN and FN are supplied explicitly. Upper95 is the exact one-sided 95% Clopper–Pearson FPR bound. The prespecified test gates use observed FPR, not that upper bound. The bounds describe the binomial calculation and do not account for residual dependence among URLs sharing a domain.

Table 4.2. Primary detection counts and rates at unchanged validation-selected thresholds.

Table 4.2a. Confusion counts; P and N are the declared stratum denominators.

Population / model	**TP/P**	**FN**	**FP/N**	**TN**
Internal: Length-only	4958/14326	9368	119/20267	20148
Internal: Logistic-L1	14137/14326	189	157/20267	20110
Internal: Transformer	14229/14326	97	205/20267	20062
Internal: Fixed cascade	14137/14326	189	157/20267	20110
External: Length-only	16/69	53	236/2497	2261
External: Logistic-L1	69/69	0	2261/2497	236
External: Transformer	69/69	0	2479/2497	18
External: Fixed cascade	69/69	0	2261/2497	236
External: GMM policy	69/69	0	2479/2497	18

Table 4.2b. Observed rates and historical row-wise binomial upper bounds. The latter do not establish population coverage for dependent URLs.

Population / model	Recall	FPR	FPR upper95
Internal: Length-only	34.6084%	0.5872%	0.6834%
Internal: Logistic-L1	98.6807%	0.7747%	0.8838%
Internal: Transformer	99.3229%	1.0115%	1.1349%
Internal: Fixed cascade	98.6807%	0.7747%	0.8838%
External: Length-only	23.1884%	9.4513%	10.4699%
External: Logistic-L1	100.0000%	90.5487%	91.4961%
External: Transformer	100.0000%	99.2791%	99.5336%
External: Fixed cascade	100.0000%	90.5487%	91.4961%
External: GMM policy	100.0000%	99.2791%	99.5336%

Structural features produce a substantial recall gain over length alone: 64.0723 percentage points internally and 76.8116 on gold positives (Table 4.3). The domain-clustered and corrected independent-pair intervals exclude zero under their respective assumptions. Internal structural FPR stays below 1%. On certified external negatives, FPR is 90.5487%, compared with 9.4513% for length-only. Measuring both classes shows the specificity cost alongside the additional detections. This comparison spans changes in source composition and representation, so it cannot separate their effects. Section 4.5 examines ranking and calibration.

Table 4.3. All six primary paired recall contrasts after the uncertainty correction. Differences and intervals are percentage points; methods and dependence assumptions are specified in Section 3.8.

Positive stratum / contrast	Candidate / reference TP	Rows / domains	Difference (pp)	95% interval (pp)
gold.cascade minus logistic l1	69 / 69	69 / 69	0.0000	[−6.1533, 6.1533]
gold.cascade minus transformer	69 / 69	69 / 69	0.0000	[−6.1533, 6.1533]
gold.logistic l1 minus length only	69 / 16	69 / 69	76.8116	[57.2709, 87.1857]
gold.policy minus cascade	69 / 69	69 / 69	0.0000	Not established
internal.cascade minus logistic l1	14137 / 14137	14326 / 9757	0.0000	Not established
internal.logistic l1 minus length only	14137 / 4958	14326 / 9757	64.0723	[59.9897, 67.6302]

Calibration selected an inclusive band with zero half-width. It contains three validation ties, but none among the 34,593 internal or 8,701 external evaluation records. Consequently, the fixed cascade never calls the transformer and exactly matches Logistic-L1, adding no recall. The corrected gold-positive interval is [−6.1533, 6.1533] percentage points; the degenerate internal bootstrap supplies no population lower bound. Neither establishes strict improvement. The inactive gate does not show that character information is absent. Transformer-only internal recall is 99.3229%, compared with

Logistic-L1's 98.6807%, although transformer FPR is 1.0115% internally and 99.2791% on certified external negatives.

RQ1 measures what the richer representation adds and opens a second question: which signals support that gain? Validation format–class association and the fixed-weight scheme test show dependence on supplied URL spelling. The different partitions do not quantify the HTTPS-attributable share of the recall gain. The zero-call cascade reproduces structural-only recall, making expert corrections a separate design question. H1 evaluates both external specificity and incremental cascade benefit.

4.3 RQ2: Monitoring and Future-Only Routing

All six GMM candidates converge, and their training Bayesian information criterion (BIC) values are reported in Table 4.4. The minimum selects six components without using external outcomes. Calibration uses 16,325 validation rows and audit uses 16,370. Each stream contains 14,783 domains and 252 complete windows, with no domains shared between streams. At the fixed boundary of −67.45792380813624, the audit flags 28/252 windows, or 11.1111%. Relative to the 5% limit, this identifies calibration-to-audit stability as a separate requirement for the selected monitor before external evaluation begins.

Table 4.4. Training-only GMM model-selection record.

Components	Training BIC	Iterations	Converged
1	12267201.154216	2	True
2	−11632435.873684	10	True
3	−20139406.839497	11	True
4	−20182632.805988	11	True
5	−18178918.655785	12	True
6	−22536770.082576	13	True

The external stream supplies 132 complete 256-row windows at stride 64. GMM alerts on 115, or 87.1212%, exceeding the 80% target. This is evidence of external P(X) departure from the fitted reference, without independently labeled harmful drift. An alert ending at t affects only t+1 through t+256. Overlapping activations are combined and truncated at the stream's end. Incomplete terminal windows do not enter the window denominator, although earlier alerts can still cover their requests.

Following an alert into the next decisions reveals what escalation contributes. The policy sends 8,253/8,701 external requests to the transformer, or 94.8512%, compared with none for the fixed band. Both detect 69/69 gold positives. Certified false positives rise from 2,261 to 2,479: an additional 218 errors, or 8.7305 percentage points. Recognizing input change therefore needs a separate test of the expert selected to respond. These results describe ordered replay; its stateful, nonrandomized design and degenerate historical bootstrap do not establish a population or randomized causal effect.

Table 4.5 compares the GMM with maximum mean discrepancy (MMD) and population stability index (PSI) diagnostics.

Table 4.5. Frozen GMM and descriptive comparator monitors. All rates use complete overlapping windows, without independent-binomial intervals.

Monitor	Fixed boundary	Original audit alerts	External alerts
GMM	−67.45792380813624	28/252 (11.1111%)	115/132 (87.1212%)
MMD	0.01662655786449553	8/252 (3.1746%)	132/132 (100.0000%)
PSI	0.12001239829542137	11/252 (4.3651%)	132/132 (100.0000%)

The comparison makes monitoring and response two measurable parts of the design. External-window sensitivity and audit false alerts describe the monitor; policy FPR and recall improvement describe the action it triggers. MMD and PSI flag every external window with fewer audit alerts, making them informative monitoring comparators. Whether an alert leads to a better decision still requires paired outcomes. The supplement retains all 396 external monitor-window records and 3,432 PSI feature scores.

4.4 RQ3: Real-HTTP Cost and Inline Viability

The HTTP tests measure the caller's full wait across the request path, exposing costs beyond isolated model scoring.

The measurements were taken on an Apple M4 Max MacBook Pro with 16 central processing unit (CPU) cores, 128 gigabytes (GB) of memory and macOS 15.6.1. The frozen runtime uses singleton transformer inference and records numerical-library

limits. Hardware and environment bindings are retained for each accepted segment. The harness times serialization, queueing, model execution, body reading and response validation. Client-phase throughput and latency exclude warmup, loading, controls and the separately reported drain. Because closed-loop timing omits unsent backlog, these results cannot establish open-loop production capacity.

The fixed gate meets the expert-invocation and primary request-error limits. Its designated invocation cell records zero physical transformer attempts across 10,000 measured requests, within the ≤ 30% limit and consistent with zero fixed-band selections. Primary cells 21–25 record 0/50,000 errors, satisfying < 0.1%, while pooled p95 across 50,000 terminal latencies is 364.3004101 ms against the 200 ms requirement. These measurements separate computation saved from time spent serving a request. Table 4.6 reports the operational groups; Table 4.7 separates throughput from post-phase drain. External detection is assessed alongside these service results.

Table 4.6. All 25 operational groups. Each row pools five complete repeats. Fixed and transformer rows contain 50,000 requests; live shift contains 43,505. The physical-attempt rate is the number of transformer forward attempts divided by all measured client requests; it is not a latency reduction.

Workload / concurrency	p50 ms	p95 ms	p99 ms	Errors / requests	Physical attempts
Fixed 1% / c1	1.274	1.489	1.694	0/50000	0.0000%
Fixed 1% / c8	5.754	10.026	13.461	0/50000	0.0000%
Fixed 1% / c16	18.482	45.579	65.890	0/50000	0.0000%

Workload / concurrency	p50 ms	p95 ms	p99 ms	Errors / requests	Physical attempts
Fixed 1% / c32	43.103	130.176	196.796	0/50000	0.0000%
Fixed 1% / c64	103.386	364.300	581.587	0/50000	0.0000%
Fixed 1% / c128	237.491	897.336	1452.179	120/50000	0.0000%
Fixed 0.1% / c1	1.264	1.480	1.646	0/50000	0.0000%
Fixed 0.1% / c8	5.767	10.037	13.618	0/50000	0.0000%
Fixed 0.1% / c16	18.703	45.351	65.702	0/50000	0.0000%
Fixed 0.1% / c32	45.833	138.859	212.348	0/50000	0.0000%
Fixed 0.1% / c64	102.178	341.839	528.288	0/50000	0.0000%
Fixed 0.1% / c128	222.653	795.941	1217.360	36/50000	0.0000%
Fixed 5% / c1	1.272	1.487	1.654	0/50000	0.0000%
Fixed 5% / c8	5.926	10.310	13.935	0/50000	0.0000%
Fixed 5% / c16	18.136	44.861	68.399	0/50000	0.0000%
Fixed 5% / c32	45.369	137.899	207.373	0/50000	0.0000%
Fixed 5% / c64	102.126	338.414	518.365	0/50000	0.0000%
Fixed 5% / c128	227.824	831.311	1291.329	59/50000	0.0000%
Transformer 1% / c1	7.142	8.084	8.581	0/50000	100.0000%

Workload / concurrency	p50 ms	p95 ms	p99 ms	Errors / requests	Physical attempts
Transformer 1% / c8	49.032	52.725	54.584	0/50000	100.0000%
Transformer 1% / c16	98.300	104.864	107.313	0/50000	100.0000%
Transformer 1% / c32	200.785	243.880	266.039	0/50000	100.0000%
Transformer 1% / c64	412.498	476.335	1017.011	74/50000	99.8800%
Transformer 1% / c128	851.982	970.861	2241.199	1612/50000	96.9640%
Live shift external / c1	10.267	11.582	12.422	0/43505	94.8512%

Table 4.7. Five-run throughput and post-phase drain ranges. Full individual run values and counters are in operational-runs.csv; ranges are descriptions, not uncertainty intervals or best-run selection.

Workload / concurrency	Attempts/s range	Successes/s range	Drain ms range
Fixed 1% / c1	758.45–771.82	758.45–771.82	27.533–31.584
Fixed 1% / c8	1243.93–1272.10	1243.93–1272.10	27.903–29.019
Fixed 1% / c16	725.04–747.06	725.04–747.06	28.040–28.777
Fixed 1% / c32	576.54–595.27	576.54–595.27	27.993–30.495
Fixed 1% / c64	398.91–502.69	398.91–502.69	27.713–30.146
Fixed 1% / c128	308.03–453.34	305.38–453.29	28.497–29.621
Fixed 0.1% / c1	763.33–774.34	763.33–774.34	27.876–29.034
Fixed 0.1% / c8	1244.78–1260.67	1244.78–1260.67	28.355–29.591
Fixed 0.1% / c16	699.52–750.18	699.52–750.18	28.396–28.985
Fixed 0.1% / c32	528.15–582.76	528.15–582.76	28.546–31.158
Fixed 0.1% / c64	435.98–530.84	435.98–530.84	28.656–30.322
Fixed 0.1% / c128	400.75–455.73	400.27–455.60	28.421–30.978
Fixed 5% / c1	761.24–771.45	761.24–771.45	26.976–28.578
Fixed 5% / c8	1172.44–1236.35	1172.44–1236.35	27.884–29.053

Workload / concurrency	Attempts/s range	Successes/s range	Drain ms range
Fixed 5% / c16	727.63–752.44	727.63–752.44	28.150–29.277
Fixed 5% / c32	514.84–590.11	514.84–590.11	27.096–29.080
Fixed 5% / c64	455.18–507.15	455.18–507.15	28.340–30.497
Fixed 5% / c128	394.32–446.43	392.67–446.30	27.862–31.244
Transformer 1% / c1	140.08–141.29	140.08–141.29	27.084–28.343
Transformer 1% / c8	162.21–163.13	162.21–163.13	27.221–28.556
Transformer 1% / c16	162.35–162.90	162.35–162.90	27.360–30.154
Transformer 1% / c32	141.74–163.30	141.74–163.30	28.055–30.852
Transformer 1% / c64	141.00–155.24	140.76–155.04	30.785–32.682
Transformer 1% / c128	136.12–150.23	133.22–145.22	36.661–38.075
Live shift external / c1	97.70–100.76	97.70–100.76	23.550–25.804

Higher concurrency does not consistently increase throughput. For the 1% fixed workload, concurrency 8 produces about 1,244–1,272 attempted requests/s, while concurrency 64 produces 399–503. At concurrency 128, the fixed workload records 120 errors at 1%, 36 at 0.1% and 59 at 5%, each among 50,000 requests. Transformer-only records 74 errors at concurrency 64 and 1,612 at 128, with pooled p95 near 476.34 and 970.86 ms. Its physical-attempt fractions below one arise from the accounting of requests, admissions and failures, rather than selective execution. All failures stay in the

denominator. Separate counters identify admitted, completed, failed and successfully scored work.

The five serialized live-shift runs reproduce the offline trace exactly. They record 41,265 physical attempts across 43,505 measured requests, or 94.8512%, with zero errors and pooled p95=11.5816 ms. These runs use concurrency one, whereas H3 uses 64. Their workload, denominator and original order also differ from the fixed reference. The agreement verifies execution of the trace without establishing primary-gate compliance or production safety.

The joint measurements show why invocation, detection and latency need their own tests. The cascade and transformer exceed the certified FPR limit, flag all 1,163 label-free Tranco controls and detect the same 69 gold positives. Under independent-pair assumptions, the corrected 95% recall-difference envelope is [−6.1533, 6.1533] percentage points. It leaves the −2-point margin unresolved and does not demonstrate inferiority. RQ3 measures low transformer use and primary request reliability, while specifying the constraints on specificity, recall uncertainty and latency. H3 satisfies two of its eight gates; the remaining six define what the complete gateway must additionally demonstrate.

4.5 Secondary Analyses and Interpretation

The secondary analyses examine discrimination, calibration and sensitivity to prevalence for every declared model, using the complete comparator set. Across one internal and six external populations, the 21 internal and 22 external detector columns produce 153 population–detector records.

Full mixed-class metrics are available for 43 records: 21 internal and 22 gold-plus-certified. These produce 430 calibration bins, 43 low-FPR curves and 129 prevalence projections. Gold and certified records retain separate confusion strata. Silver and bronze results report positive recall by source and tier, while Tranco reports only label-free alerts.

Single-class strata report only measures supported by their labels. Average precision (AP), receiver operating characteristic area under the curve (ROC AUC) and calibration remain undefined where they cannot be calculated. Table 4.8 displays AP, ROC AUC, Brier score and expected calibration error (ECE) for the selected mixed-class comparisons. A source-only classifier is unavailable because PhiUSIIL lacks per-record provenance independent of class. Inferring source from outcome would be circular, so the formatting and transfer analyses do not isolate collection-source effects.

Table 4.8. Selected full-metric summaries. External AP and calibration describe the observed 69-positive/2,497-negative mixture, not production prevalence. All columns, including precision, F_2, MCC, and balanced accuracy, remain in secondary-metrics.csv.

Population / detector	**AP**	**ROC AUC**	**Brier**	**10-bin ECE**
Internal: Length-only	0.736846	0.712485	0.195827	0.116636
Internal: Logistic-L1	0.995978	0.994856	0.008240	0.009815
Internal: Transformer	0.998310	0.998042	0.003707	0.000514
Internal: Fixed cascade	0.995978	0.994856	0.008240	0.009815
Internal: Formatting	0.768122	0.802108	0.131705	0.052188

Population / detector	AP	ROC AUC	Brier	10-bin ECE
Internal: Random Forest	0.996749	0.996394	0.004519	0.002926
Gold + certified: Length-only	0.218936	0.383893	0.323470	0.517273
Gold + certified: Logistic-L1	0.314882	0.851239	0.876785	0.879966
Gold + certified: Transformer	0.099927	0.592273	0.965747	0.965905
Gold + certified: Fixed cascade	0.314882	0.851239	0.876785	0.879966
Gold + certified: Formatting	0.026890	0.494994	0.103375	0.268231
Gold + certified: Random Forest	0.036487	0.634068	0.867826	0.879992
Gold + certified: GMM policy	0.175659	0.498204	0.965747	0.965902

Logistic-L1 average precision falls from 0.995978 internally to 0.314882 externally. External ECE is 0.879966 for that model and about 0.9659 for the transformer and policy. Above-chance ranking does not make the frozen threshold useful. The descriptive low-FPR curves enumerate tied, uninterpolated cutoffs without selecting new deployment thresholds. Near 0.999992, the structural curve detects 20/69 positives with 24/2,497 false positives. Because that point was inspected retrospectively, it cannot replace the frozen result of 2,261 false positives.

The prevalence projections apply measured class-conditional true-positive rate (TPR) and FPR to hypothetical phishing mixtures of 0.1%, 1% and 5%. They are separate from HTTP measurements. At 1%, the external fixed-cascade rates imply about 9,064 alerts per 10,000 requests, including 8,964 false alerts and no misses. This shows the burden associated with poor specificity under the assumed mixture. It verifies neither

deployment prevalence nor transfer of those rates to another population. All rates and projections are retained. Table 4.9 separates positive tiers and label-free controls so that alert rates are not mistaken for mixed-class performance.

Table 4.9. Additional positive tiers and label-free controls at frozen primary thresholds.

Detector	**Silver TP / 417**	**Bronze TP / 4,555**	**Tranco alerts / 1,163**
Length-only	18 (4.3165%)	35 (0.7684%)	0 (0.0000%)
Logistic-L1	417 (100.0000%)	4555 (100.0000%)	1163 (100.0000%)
Transformer	417 (100.0000%)	4554 (99.9780%)	1163 (100.0000%)
Fixed cascade	417 (100.0000%)	4555 (100.0000%)	1163 (100.0000%)
GMM policy	417 (100.0000%)	4554 (99.9780%)	1163 (100.0000%)

The five-indicator formatting comparator reaches 60.4216% internal recall with no observed false positives. Externally, it detects no gold positives and has 1.0012% certified FPR. This restricted representation shows source-dependent structure without proving a causal shortcut. The fixed 100-tree Random Forest achieves internal recall of 99.2042%, FPR of 0.7006%, AP of 0.996749 and AUC of 0.996394. Externally, the corresponding values are 100%, 91.3496%, 0.036487 and 0.634068. Its corrected artifact and 0.2 validation cutoff remain secondary evidence. Table 4.10 presents every accepted historical permutation comparator, whose provenance limits are discussed next.

Table 4.10. Every accepted permutation comparator; no seed selection.

Permutation seed	Internal AP	Internal AUC	External AP	External AUC
42	0.533318	0.481551	0.025910	0.369989
43	0.672760	0.624953	0.272610	0.827396
44	0.551078	0.546285	0.017823	0.176574
45	0.437844	0.402251	0.128390	0.837681
46	0.772821	0.687614	0.190236	0.738530

The historical controls lack digests linking their reconstructed permutations to the labels consumed by fitting. Section 4.10.4 supplies audited controls and a mechanism analysis of their ranking variation. Those checks explain the audited pipeline without authenticating historical estimators. Table 4.11 gives the separate transformer seed/runtime sensitivity.

Table 4.11. Transformer seed/runtime sensitivity at each accepted secondary operating point. T denotes transformer. All five secondary cascades coincide with Logistic-L1 on these evaluated populations.

Seed	Internal T recall	Internal T FPR	Gold T recall	Certified T FPR	Fixed band selections
42	99.3229%	1.0115%	100.0000%	99.2791%	0 internal; 0 external
43	99.3159%	0.7253%	100.0000%	99.3192%	0 internal; 0 external
44	99.3369%	0.8783%	100.0000%	99.3192%	0 internal; 0 external
45	99.3299%	1.0362%	100.0000%	99.3592%	0 internal; 0 external
46	99.3020%	0.7549%	100.0000%	99.3192%	0 internal; 0 external

All seeds 42–46 are reported as one accepted family. Seed 42 uses the historical runtime, and seeds 43–46 use the later pinned runtime, so their differences cannot isolate the seed effect. Each model makes three logical band selections among 32,695 validation rows and none in either evaluation stream. All cutoffs, bands, checkpoints and development values are retained in complete-secondary-results.json; no best seed is selected for promotion. Table 4.12 provides the supplementary positive-only paired tests and multiplicity adjustment.

Table 4.12. Positive-only paired McNemar tables and four-slot Holm adjustment. "Underflow" means the saved floating-point computation returned 0.0, not an exact mathematical p-value of zero.

Contrast	Both correct	Candidate only	Reference only	Both incorrect	Exact p	Holm p
internal logistic minus length	4952	9185	6	183	underflow	underflow
internal cascade minus logistic	14137	0	0	189	1	1
external gold logistic minus length	16	53	0	0	2.22045e-16	6.66134e-16
external gold cascade minus logistic	69	0	0	0	1	1

The nominal tests compare aligned positive outcomes without modeling domain or routing dependence. They supplement the clustered contrasts and do not determine them. Gateway utility requires recall improvement and external specificity to be assessed together. Table 4.13 examines input sensitivity through label-free probes, which do not measure detection accuracy.

Table 4.13. Accepted label-free development probe accounting.

Stream	Rows	Eligible	Changed	Eligible no-op	Ineligible
original	16370	16370	0	16370	0
ascii scheme host uppercase	16370	16370	16370	0	0
percent escape hex uppercase	16370	53	22	31	16317
first literal path alphanumeric percent encode	16370	1563	1563	0	14807

Each probe stream contains 16,370 positions and 252 complete overlapping windows. Uppercasing American Standard Code for Information Interchange (ASCII) letters in the scheme and host changes no binary decisions. Monitor alerts change from 28→29 for GMM, 8→15 for MMD and 11→10 for PSI. Uppercasing percent escapes affects 22 eligible records without changing decisions or monitor alerts, providing only limited evidence for broader invariance. Encoding the first literal path character affects 1,563 records. Under that probe, 105 GMM-policy decisions change 0→1 and 63 change 1→0. Length-only adds 48 alerts and transformer adds 14; fixed cascade and Logistic-L1 add none. GMM, MMD and PSI window alerts become 213, 14 and 6. The files probe-decisions-and-scores.csv and probe-monitors-and-scores.csv preserve paired score changes, all four decision cells, monitor transitions, fixed boundaries and eligibility

counts. These probes inherit no outcome labels and establish neither accuracy nor semantic equivalence, adversarial success or production robustness.

4.6 Evaluation Provenance and Information Exposure

Three features of the evaluation record define its interpretive scope: the amendment to external input spelling, completion of the service matrix across sessions, and analyst exposure to test records. Appendix C retains the detailed history and failed predecessors. The points below connect each feature to the comparisons it affects.

The initial raw-field preparation admitted no gold or Tranco cases because their publisher strings lacked schemes. The amendment used exact publisher url_norm for parsing and model input, preserving raw values, labels, overlap rules and thresholds. It followed label-count inspection and preceded amended prediction, so the choice was informed by preparation. Appendix C records the full amendment.

The operational matrix includes all 72 eligible prefix cells and the 53-cell continuation, excluding partial requests and earlier substitutes. Section 3.12 defines its measurement units; Appendix C preserves the complete execution record.

Two sessions were needed to complete the planned single-session matrix. Descriptive group 71–75 crosses their boundary; primary cells retain their original measurements. Thermal state, caches, background activity and elapsed time may change with the fixed workload order. Session labels cannot separate those effects, and environmental checks were sampled rather than continuous. Repeated scoring and broad searches on September 3 and September 9 also exposed test rows to the analyst. Models and thresholds stayed frozen, but the population was neither lifetime-unseen nor evaluated in a single lifetime pass.

4.7 Complete Hypothesis Adjudication

Table 4.14 evaluates the original combined operating requirements. Its rows sometimes reuse the same measurement, so the row count is not a count of independent findings. Recall rows report lower confidence endpoints, with point estimates in Table 4.3. Dependence or a degenerate bootstrap can leave a required lower bound unestablished. Table 4.2 supplies counts and historical binomial bounds, while corrected-primary-gates.csv retains the full-precision decisions (Appendix A). The H3 latency and error checks use the five designated primary runs, rather than all 125 cells.

Both the original and corrected decisions are retained in corrected-primary-gates.csv; primary-gates.csv preserves the original analysis. The historical row-wise binomial bounds remain subject to the dependence limitations in Section 3.8. All FPR thresholds are unchanged.

Table 4.14. All 22 primary hypothesis checks with unchanged numerical requirements and corrected paired inference. A failed inferential check means its requirement is not established.

H	Condition	Observed value	Required	Result
H1	Internal: length-only FPR	0.5872%	≤ 1.0000%	pass
H1	Internal: logistic L1 FPR	0.7747%	≤ 1.0000%	pass
H1	Internal: cascade FPR	0.7747%	≤ 1.0000%	pass
H1	Certified negatives: length-only FPR	9.4513%	≤ 1.0000%	not met
H1	Certified negatives: logistic L1 FPR	90.5487%	≤ 1.0000%	not met
H1	Certified negatives: cascade FPR	90.5487%	≤ 1.0000%	not met
H1	Internal recall: logistic L1 minus length-only	59.9897 pp	> 0.0000 pp	pass
H1	Internal recall: cascade minus logistic L1	Not established	> 0.0000 pp	not met
H1	Gold recall: logistic L1 minus length-only	57.2709 pp	> 0.0000 pp	pass
H1	Gold recall: cascade minus logistic L1	−6.1533 pp	> 0.0000 pp	not met
H2	External-window alert rate	87.1212%	≥ 80.0000%	pass
H2	Audit-window alert rate	11.1111%	≤ 5.0000%	not met

H	Condition	Observed value	Required	Result
H2	Certified negatives: policy FPR	99.2791%	≤ 1.0000%	not met
H2	Gold recall: policy minus cascade	Not established	> 0.0000 pp	not met
H3	Certified negatives: cascade FPR	90.5487%	≤ 1.0000%	not met
H3	Certified negatives: transformer FPR	99.2791%	≤ 1.0000%	not met
H3	Tranco: cascade alert rate	100.0000%	≤ 1.0000%	not met
H3	Tranco: transformer alert rate	100.0000%	≤ 1.0000%	not met
H3	Gold recall: cascade minus transformer	−6.1533 pp	≥ −2.0000 pp	not met
H3	Reference transformer invocation rate	0.0000%	≤ 30.0000%	pass
H3	HTTP pooled p95 latency	364.3004 ms	≤ 200 ms	not met
H3	HTTP request-error rate	0.0000%	< 0.1000%	pass

The 22 checks give a complete accounting of the gateway's operating requirements: H1 meets five of 10 conditions, H2 one of four and H3 two of eight, for eight passing component checks. The joint hypotheses are unsupported because each requires every condition. Corrected uncertainty changes one component decision: the original 69-pair comparison does not establish the two-point recall noninferiority margin. The following comparisons use the measured structural gain, paired-error patterns and service observations to identify and evaluate specific design interventions.

4.8 Diagnostic Investigation and Detection Comparison

4.8.1 From Aggregate Behavior to Testable Mechanisms

Each diagnostic tests an explanation for observed behavior. Changing scheme spelling with weights fixed tests input sensitivity. Paired model errors establish available expert corrections. Removing model work from the HTTP control probes delay elsewhere in the request path. These comparisons identify changes to implement and evaluate.

The supplied URL format provides a concrete clue. All evaluated internal negatives have is_https=1 and zero path length; external gold-positive inputs have is_https=0. These summaries concern the stated subsets and publisher url_norm, not every training row or live transport. Validation format–class association in Section 4.10.4 complements the fixed-weight test in Section 4.10.1. A prefix changes several predictors, including is_https and string composition, so removing scheme dependence requires more than dropping one indicator. The evidence does not identify how collection practices produced the association. Separately, the no-model control motivates changing client organization while keeping the scorer fixed.

The paired decisions give an exact limit on what routing can do. Both models falsely flag the same 2,261 of 2,497 certified negatives. The transformer repairs none of those errors, adds 218 false positives and shares only 18 correct negatives with the structural model. Even a router that chooses correctly whenever either model does must retain every shared false positive. Its FPR floor is 2,261/2,497, or 90.5487%. For these fixed decisions, improving the gate cannot lower that floor. Changed thresholds, additional evidence or retraining would create different decisions and require a new evaluation.

The ranking results help explain why changing the operating point is a separate question. Structural ROC AUC is 0.994856 internally and 0.851239 on gold-plus-certified external data; the transformer values are 0.998042 and 0.592273. At the best retrospectively inspected threshold with observed certified FPR at most 1%, the structural model detects 20/69 gold positives and the transformer detects 7/69. Thus, the inspected thresholds cannot retain perfect observed gold recall at the intended FPR. Since outcome labels informed these frontiers, they remain diagnostic rather than validated deployment thresholds.

The fixed cascade's inactivity follows from calibration: its width is zero, and evaluation contains no ties (Section 4.2). Replaying 43,294 saved feature vectors, routing memberships and decisions reproduces the accepted outputs. With that behavior verified, wider gates can be examined against the original rule and its recorded decisions.

Exploratory wider-gate development reuses exposed internal group-test outcomes. Before replay, the specification fixes symmetric, below-threshold and above-threshold widths of 0.025, 0.05, 0.10 and 0.20. Candidates must keep FPR at most 1%, recall within two points of transformer-only and calls at most 30%. Selection first maximizes detections, then minimizes false positives and calls, followed by the declared width/name tie-breaks. One symmetric and one directional candidate remain. These internal results cannot provide independent confirmation because they informed selection. Section 4.10.2 evaluates the selected rules on a separate cohort.

The selected symmetric band, with half-width 0.20, sends 1,146/34,593 internal records to the expert, or 3.3128%. Relative to structural-only, true positives rise from 14,137 to 14,171 and false positives fall from 157 to 117. An equal-call hash-ranked

control produces 14,142 true positives and 166 false positives. The below-threshold band uses 1,039 calls and produces 14,172 true positives and 199 false positives. On the exposed external cohort, both gates still detect 69/69 gold positives, while certified false positives rise from 2,261 to 2,272. The additional decisions comprise 11 harms and no repairs. These counts measure routing rather than timing, and the finite-batch hash control is not an online policy. The contrasting internal and external outcomes identify source-specific expert corrections as the relevant test for widening the gate.

4.8.2 Population Flow and Operating-Point Comparison

Table 4.15 traces admission to the additional benchmark. The 2,808 quarantined rows can have more than one reason among the four exclusion categories, so reason incidences cannot be added to obtain the row total. All 8,622 eligible rows receive predictions in source order. Fitting and external scoring complete successfully for the entire admitted population. Separate arithmetic reproduces every saved metric, interval and D decision without refitting or making new predictions.

Table 4.15. Additional benchmark admission; reason incidences are overlapping.

Item	Count	Interpretation
Publisher rows	11,430	5,715 per reference class
Quarantined rows	2,808	Unique excluded rows
PhiUSIIL domain overlap	2,134	Reason incidence, before filtering
PhishVN domain overlap	1,212	Reason incidence, before filtering
Invalid URL	108	Reason incidence
Same-label canonical duplicate	10	Reason incidence
Eligible publisher-phishing	4,651	2,897 registrable domains
Eligible publisher-legitimate	3,971	3,377 registrable domains
All eligible rows	8,622	6,273 domains; largest cluster 351 rows

The refit removes an input dependency and quantifies its detection trade-off. At the two frozen operating points in Table 4.16, it produces 251 fewer false positives and 184 fewer true positives. Paired labels make both consequences of the invariance repair explicit.

Table 4.16. Additional benchmark detection results; P=4,651 and N=3,971.

Quantity	Unchanged 25-feature model	Transport-neutral 24-feature model
Threshold	0.2670846328	0.6815549263
TP / FN	4,619 / 32	4,435 / 216
FP / TN	3,940 / 31	3,689 / 282
Recall	99.31%	95.36%
FPR	99.22%	92.90%
Precision	53.97%	54.59%
ROC AUC	0.6821	0.6950
Average precision	0.6651	0.6976
Brier score	0.4306	0.4077

The candidate-minus-comparator recall difference is −3.9561 percentage points. Its 97.5% domain-cluster interval is [−5.0547, −3.0014] (Figure 4.1), while the candidate FPR interval is [91.8651%, 93.9100%]. Neither interval has undefined resamples among the 10,000 replicates. The largest cluster contains 351 rows; treating all 8,622 rows as independent would ignore this dependence. The ROC AUC, average-precision and Brier differences in Table 4.16 are descriptive. Figure 4.2 shows calibration together with the number of observations in each bin, making sparse bins visible.

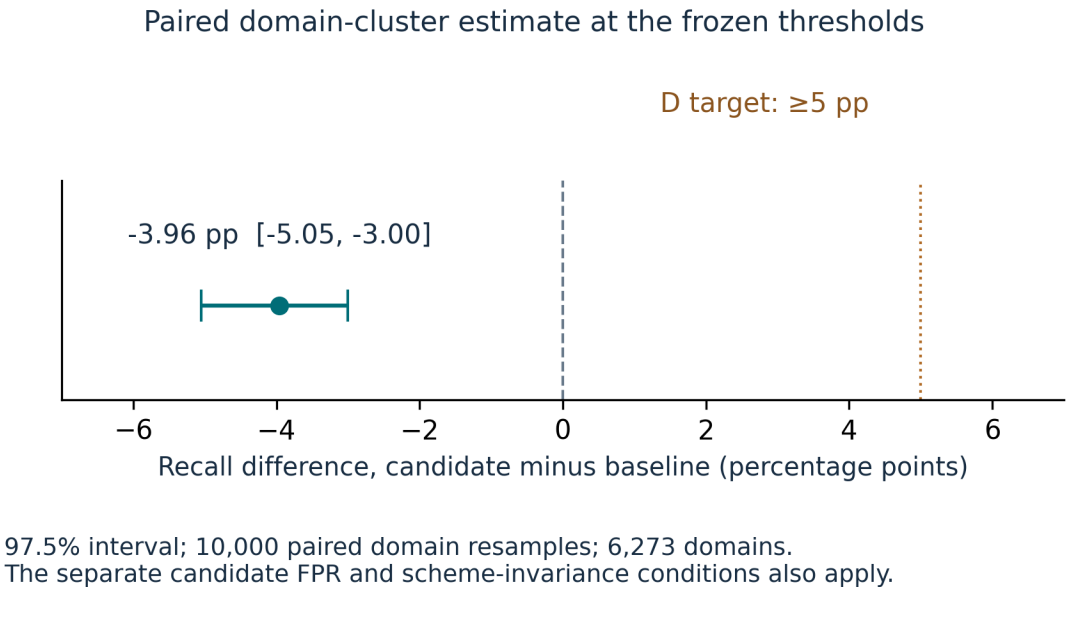


Figure 4.1. Candidate-minus-comparator recall difference at frozen thresholds; 97.5% domain-cluster interval.

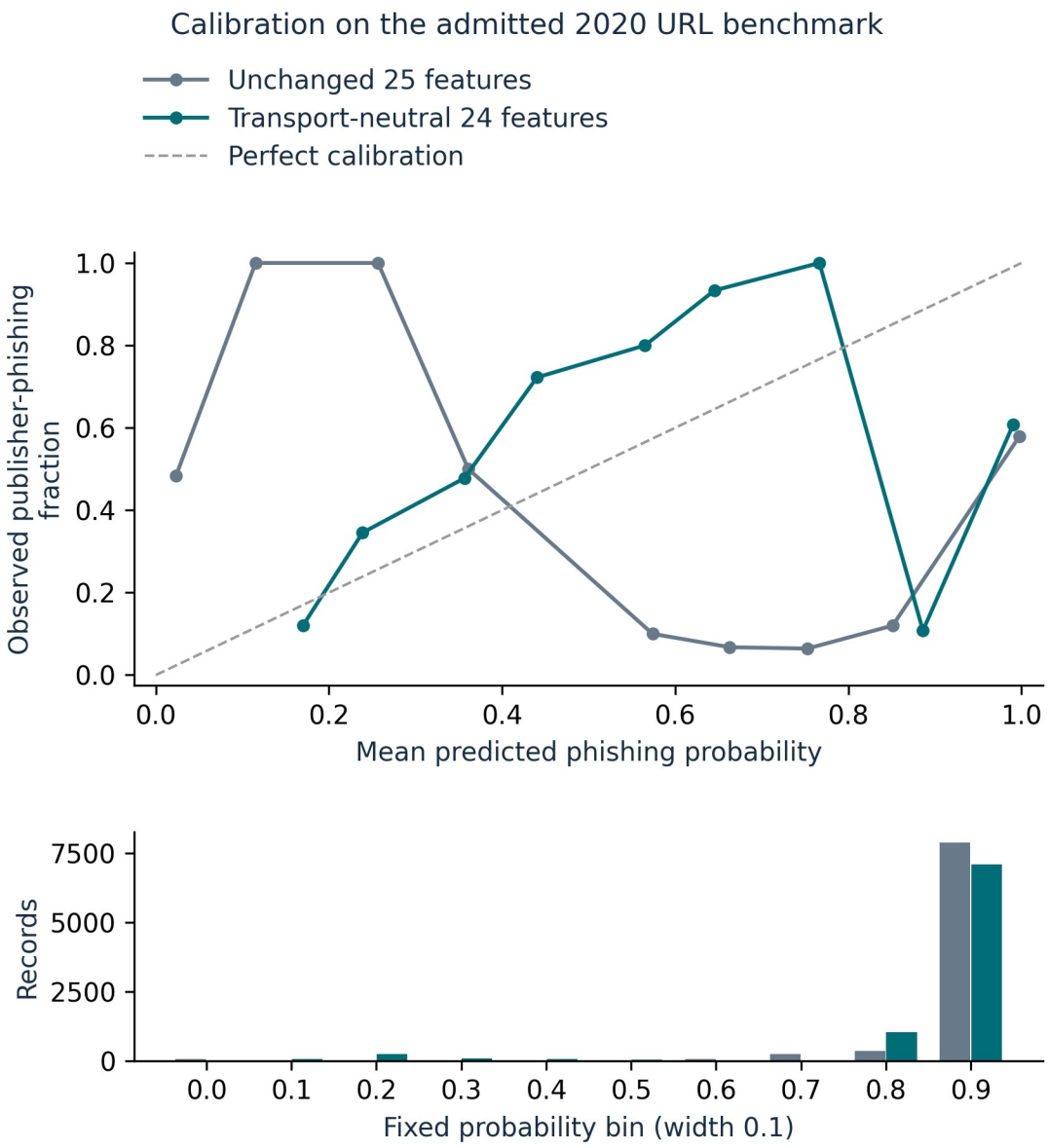


Figure 4.2. Fixed-bin calibration and bin populations on the admitted benchmark.

Sparsely populated bins can have extreme fractions; empty bins have no reliability point.

The class mixture is not deployment prevalence.

4.8.3 Representation Invariance and Decision

The refitted representation removes scheme sensitivity exactly across the tested population. All 8,622 candidate feature vectors and scores match their opposite-scheme companions, with no decision changes (Table 4.17). The comparator has 2,909 exact score matches, 481 decision flips and a maximum absolute score change of 0.99665. A flip shows sensitivity; whether it is an error requires a label. The companions add no independent outcome labels.

Table 4.17. Scheme-neutral representation D: requirements and decisions.

Requirement	Observed result	Decision
Recall gain at least 5 pp	−3.9561 pp	Not met
97.5% recall-difference lower bound above zero	−5.0547 pp	Not met
Candidate observed FPR at most 1%	92.8985%	Not met
Exact candidate feature/score invariance	8,622 / 8,622; zero decision flips	Met
D conjunction	Three efficacy requirements not met	Not supported

D verifies scheme invariance. The combined normalization, refitting and recalibration also reduce both false positives and true-positive detections, with FPR above the ceiling. Stability and class separation are distinct properties; the joint requirement tests both.

4.9 Controlled Service Comparison: Completed Evaluation

4.9.1 Workload Scope and Complete Accounting

The final service schedule contains 80 arms, crossing two workloads and two concurrency levels with 10 pairs and two client topologies. It produces 800,000 measured requests and 80,000 excluded warmups, with all 80 processes exiting normally. The interrupted predecessor remains separate in Appendix C. These measurements describe the specified synthetic, structural-only closed-loop workload.

The estimation set contains all 80 arms. The interrupted predecessor contributes no observations; its replacement rule was fixed before the primary structural concurrency-64 measurements.

Independent arithmetic reproduces all 80 arm summaries, 40 pairs, eight pooled groups and five S decisions. The schedule contains one shared-client timeout and no warmup errors. The original 125-cell experiment remains separate.

4.9.2 Paired Tail-Latency Effect

Giving each worker its own client reduces p95 in every primary pair (Table 4.18). The median worker/shared run ratio is 0.203249, a 79.68% median paired reduction. Across 10,000 fixed run-pair resamples, the two-sided 97.5% interval is [0.187903, 0.213508]. Every pair and resample has a defined result. The estimate describes this synthetic structural-only HTTP setup.

The no-model control helps explain where the time was saved. Pooled successful-request p95 drops from 358.3663 to 71.9990 ms with structural scoring and from 345.95 to 67.21 ms with no model. Both improvements point to the client/transport path, without isolating a particular queue or scheduling cost. Worker p95 meets the 200-ms

requirement in this setup. Pooled request quantiles and the median paired run ratio summarize different quantities; Section 4.9.3 reports the full control and response comparisons.

Table 4.18. All ten structural-detector concurrency-64 pairs; success-only p95 in milliseconds.

Pair	Shared p95	Worker p95	Ratio	Exact / requested
1	343.1552	72.6055	0.211582	85 / 10,000
2	371.7900	67.2667	0.180927	83 / 10,000
3	359.2445	70.0094	0.194879	71 / 10,000
4	358.8575	72.8568	0.203024	79 / 10,000
5	397.6750	71.5949	0.180034	89 / 10,000
6	342.2972	74.7204	0.218291	80 / 10,000
7	362.0759	74.3130	0.205242	75 / 10,000
8	352.6524	71.7555	0.203474	84 / 10,000
9	341.7835	73.6319	0.215434	71 / 10,000
10	353.7333	71.2943	0.201548	82 / 10,000

4.9.3 Controls, Throughput, Errors and Response Semantics

Table 4.19 pools ten arms and 100,000 measured attempts for each required control/concurrency combination. Attempted throughput divides attempts by the sum of

client-phase durations, excluding warmup and drain. When no errors occur, successful and attempted throughput are equal. Shared structural c64 has 99,999 successes and one error. The aggregate comma-separated values (CSV) files retain successful, failed and all-request percentiles, along with physical counters. Appendix B reports every arm.

Table 4.19. Complete pooled service groups; success p50/p95/p99 in milliseconds and attempted throughput in requests per second.

Workload / c / client	p50	p95	p99	Errors	Requests/s
Control / 1 / shared	0.93	1.10	1.24	0	1039.88
Control / 1 / worker	0.93	1.11	1.26	0	1036.52
Control / 64 / shared	103.91	345.95	516.29	0	469.21
Control / 64 / worker	35.86	67.21	77.34	0	1647.64
Structural / 1 / shared	1.17	1.42	1.63	0	820.44
Structural / 1 / worker	1.17	1.43	1.63	0	819.76
Structural / 64 / shared	107.52	358.37	536.11	1	454.14
Structural / 64 / worker	37.36	72.00	85.37	0	1526.71

At concurrency 1, client topology makes little difference to pooled p95: control values are 1.10 versus 1.11 ms, and structural-scoring values are 1.42 versus 1.43 ms. At concurrency 64, control p95 falls from 345.95 to 67.21 ms, and structural p95 falls from 358.37 to 72.00 ms. Structural attempted throughput rises from 454.14 to 1,526.71 requests/s; successful shared throughput is 454.14. The corresponding no-model

improvement points to the client/transport path. Cross-condition percentiles cannot be added to decompose request time, identify a particular queue or pool, or estimate a transformer speedup.

The only error occurs in the shared arm of structural c64 pair 2: a timeout recorded at 2,019.144208 ms. Since it is the sole failure, it supplies all three failure-only quantiles. Cancellation and bookkeeping can delay reporting beyond the two-second deadline. The request remains a failure and was never admitted; the arm admits and completes 9,999 requests. Across the full schedule, 799,999 requests are admitted and completed. There are zero server failed-request increments, transformer attempts or successful transformer scores. The timeout is a client error, which is counted separately from server failures.

All 100,000 primary worker requests succeed. The shared timeout leaves 99,999 pairs with comparable responses. When request IDs and admission sequence are excluded, each pair agrees in action, probability and stage-2 invocation.

The prespecified strict comparison excludes request IDs alone. It therefore requires admission sequence to agree as well. Only 799/100,000 primary pairs meet that requirement. Another 99,200 comparable pairs differ solely in sequence; the timeout pair is noncomparable.

Across the 40 schedule pairs, predictions agree on all 399,999 comparable requests. Of the 400,000 requested pairs, one remains noncomparable because of the timeout.

4.9.4 Requirement S and Engineering Interpretation

S meets four of its five requirements (Table 4.20), demonstrating lower latency and reliable worker requests. Comparable predictions also agree. Admission-sequence equality, the fifth requirement, is not preserved. Each threshold and required response field keeps its prespecified definition.

Table 4.20. Prespecified S requirements and verified decisions.

Requirement	Observed result	Decision
Median paired ratio ≤ 0.8	0.203249	Pass
97.5% ratio upper bound < 1	0.213508	Pass
Pooled worker success p95 ≤ 200 ms	71.9990 ms	Pass
Worker errors < 0.1%	0 / 100,000 (0%)	Pass
Exact paired responses except IDs	799 / 100,000	Not met

Worker-owned clients improve latency and reliability in this experiment, preserving every comparable prediction. Applications requiring identical admission order need a separately evaluated ordering mechanism. The single-host, synthetic, structural-only closed-loop result is separate from H3 and does not measure transformer-service acceleration.

4.10 Mechanism Tests: Representation, Routing and Cost

These tests examine decision changes and computation. Frozen-weight scheme changes isolate spelling sensitivity. The domain-balanced cohort measures revised gates'

repairs and new errors, while invocation counters record actual work. Audited shuffled-label fits explain strong validation ranking after training-label randomization.

4.10.1 Frozen-model Scheme Intervention

Changing the prefix changes decisions even when the structural model is held fixed (Table 4.21). HTTP detects all 69 gold positives and flags every certified negative. With Hypertext Transfer Protocol Secure (HTTPS), certified false positives drop from 2,261 to 50 and gold detections from 69 to 22. The change removes 2,211 false positives and 47 true-positive detections. FPR is 2.0024%, above 1%, and recall is 31.8841%. The paired intervention demonstrates scheme dependence and measures its cost in both classes. It also shows why normalization needs a labeled detection test.

Table 4.21. Frozen structural-model scheme intervention. P=69 gold positives; N=2,497 certified negatives. Every character after the scheme colon is preserved.

Input form	TP	FN	FP	TN
Retained spelling	69	0	2,261	236
Forced HTTP	69	0	2,497	0
Forced HTTPS	22	47	50	2,447

HTTPS reduces secondary-positive detections from 4,972 to 687 and label-free Tranco alerts from 1,163 to 63. F demonstrates scheme dependence with weights and threshold fixed; the validation association gives related evidence about internal separability. D asks a different question by refitting and recalibrating on the original

development partitions before evaluating another cohort. Its invariance and detection effects cannot be pooled with F or attributed to normalization alone.

4.10.2 Expanded Domain-disjoint Routing Evaluation

The separate cohort tests whether the wider gates select useful expert corrections. In Table 4.22, both revised gates produce 1,046 false positives among 1,262 negatives, or 82.8843%, compared with 1,028, or 81.4580%, for structural-only and the original cascade. The 18 added errors accompany no change in positive decisions: every system detects all 1,392 positives. Always-transformer produces 1,221 false positives, or 96.7512%. Symmetric and directional matched-count controls add only two and one, respectively. These controls are finite-batch comparisons rather than deployable routing policies.

Table 4.22. R cohort decisions and logical expert selections. P=1,392; N=1,262. Every row uses the same one-URL-per-domain cohort; calls are out of 2,654 records.

System	TP	FN	FP	TN	Calls
Structural	1,392	0	1,028	234	0
Original cascade	1,392	0	1,028	234	0
Transformer	1,392	0	1,221	41	2,654
Symmetric 0.20	1,392	0	1,046	216	21
Directional 0.20	1,392	0	1,046	216	18
Symmetric control	1,392	0	1,030	232	21
Directional control	1,392	0	1,029	233	18

The recall comparison supports preservation of positive detections at these operating points. Original and symmetric cascades share 1,392 detections with transformer-only, with no positive discordances. Under independent labeled-pair assumptions, their conservative two-sided 95% envelope is [−0.3143, 0.3143] percentage points, above the −2-point lower-margin requirement. Newcombe method 10 gives [−0.2752, 0.2752]. This is recall-only noninferiority at a detection ceiling; the false-positive counts and selected corrections determine expert utility. The positives occupy 56 source-event-date groups, the largest containing 362. Resampling all-zero differences gives descriptive [0,0], which cannot establish campaign independence or replace the nondegenerate interval.

Routing quality depends on the corrections selected as well as how often the expert is called. Symmetric routing makes 21/2,654 calls, or 0.7913%, and directional routing 18/2,654, or 0.6782%, both below 30%. Among all transformer decisions, one would repair a structural false positive and 194 would introduce one. Both gates select zero repairs, preserve recall and exceed 1% FPR. Structural ROC AUC/AP are 0.7549/0.7485; transformer values are 0.7062/0.6909. These rankings describe the near-balanced cohort. The paired decisions judge selection, while the cohort mixture supplies no estimate of deployment prevalence.

4.10.3 Measured Detector Computation

C completes all 30 arms, comprising 30,000 measured requests and 3,000 excluded warmups. For each 1,000 measured requests, symmetric routing makes 17 transformer calls, directional routing makes 15, the original cascade makes none and the symmetric matched-count control makes 17. All 33,000 requests across both phases finish without recorded failure. Transformer starts and completions each total 5,790. The measured comparisons use counter differences after warmup rather than the totals for entire runs.

Table 4.23. Actual detector computation on C's fixed 1,000-string workload. Quantiles are medians of five run quantiles in milliseconds. Ratio is the median paired summed-scoring-time ratio to structural-only, not an HTTP latency ratio.

Scoring path	Calls / 1,000	p50 ms	p95 ms	p99 ms	Time ratio
Structural	0	0.187	0.198	0.226	1.000
Original cascade	0	0.189	0.204	0.245	1.009
Symmetric 0.20	17	0.189	0.223	6.769	1.646
Directional 0.20	15	0.190	0.217	6.802	1.589
Symmetric control	17	0.189	0.233	7.015	1.710
Transformer	1000	5.559	6.657	16.484	30.781

A small invocation fraction can have a large cost in the slowest requests. Symmetric routing executes the transformer for 17 of every 1,000 inputs. Its summed scoring time is 1.646 times structural-only, with paired-run range 1.585–1.743; p99 is 6.769 ms versus 0.226 ms. Five repeats on one host support a workload-specific measurement, without a run-level confidence interval. Section 4.10.2 reports the separate detection comparison. These scoring costs cannot be combined with the synthetic HTTP client improvement to infer a net service speedup.

4.10.4 Shuffled-label Controls and Ranking Mechanism

The five audited P fits consume the specified shuffled labels, with original row order and class counts preserved. They converge below 5,000 iterations, and independent

score reconstruction reproduces Table 4.24. Its AUC/AP evaluate unchanged validation labels; its TP/FP additionally use those labels for descriptive threshold selection. The latter are selected validation outcomes, not independent null-test estimates. The following analyses test why substantial ranking remains after training-label randomization.

Table 4.24. All five audited shuffled-label fits. Validation P=12,486; N=20,209. AUC and AP use unchanged validation labels; TP/FP use validation-selected descriptive operating points.

Seed	ROC AUC	AP	TP	FP	Iterations
42	0.4718	0.4871	1,623	178	2001
43	0.6419	0.6718	4,460	175	698
44	0.5487	0.5206	1,250	176	1010
45	0.3940	0.4027	730	178	700
46	0.6965	0.7683	6,882	177	341

The original audited controls have AUCs from 0.3940 to 0.6965 and AP from 0.4027 to 0.7683. Their coefficient-vector Euclidean norms are only 0.0299–0.0851, and the central half of each seed's validation probabilities lies within approximately 0.4977–0.5024. Such small score differences can still produce a consistent ordering. AUC counts the fraction of positive–negative pairs ordered correctly, giving half credit for ties. Multiplying all nonconstant logits by a positive constant changes their magnitude while preserving that order. Probabilities near 0.5 therefore do not imply chance AUC.

The validation data retain a strong relationship between format and class that random coefficients can follow. All 20,209 negatives have is_https=1, compared with 5,823 of 12,486 positives, or 46.6362%. A score consisting only of the negative HTTPS indicator has AUC 0.7668 on these labels. Shuffling training labels breaks their original association with training features, but leaves the validation association intact. Finite-sample correlations can produce small fitted coefficients that align with this structure or run against it. The same random fitted direction orders every validation row through the same predictors. Having many rows therefore does not turn these scores into independent random guesses.

The saved coefficients and fixed-score intervention directly support that explanation. Seeds 43 and 46 assign is_https coefficients of −0.008797 and −0.009530; their AUCs are 0.6419 and 0.6965. Removing only that term lowers their AUCs to 0.4031 and 0.4195. Seed 45 assigns a positive coefficient of 0.004024: its AUC rises from 0.3940 to 0.4934 when the term is removed. The same operation changes seed 42 from 0.4718 to 0.5476 and seed 44 from 0.5487 to 0.5883. These are all five original controls. The intervention identifies a substantial contribution to ranking from this term, without claiming that it uniquely accounts for the multifeature score.

All 99 scheduled additional permutations complete without warnings and below the iteration limit. Against unchanged validation labels, AUC ranges from 0.1046 to 0.9496, with mean 0.4779, median 0.4479 and empirical 2.5th–97.5th percentiles [0.2061, 0.7718]. Of the 99 fits, 12 reach or exceed the original maximum of 0.6965. This frequency describes the scheduled randomizations; it is not a leakage p-value. The

complete larger set reproduces the broad spread, showing that the original high value need not result from choosing a favorable seed after fitting.

Two reference comparisons locate the spread. With each fitted score held fixed and validation labels independently shuffled, mean AUC is 0.5000 and the full range is [0.4933, 0.5066]. Constant scores give AUC 0.5. Across training permutations, the descriptive Spearman correlation between the HTTPS coefficient and true-label AUC is -0.8253. Removing the explicit term changes mean absolute departure from 0.5 from 0.1424 to 0.0851. Equal total weight per validation domain still gives AUCs spanning 0.0770–0.9554; repeated URLs within domains are therefore insufficient to explain the spread. These calculations retain one fixed collection and do not establish independent campaign sampling.

Stricter optimization leaves the ranking explanation intact. All five continuations finish without warnings, and the largest absolute AUC change is 0.000406. The maximum first-order residual, scaled to mean loss, is 3.48e-09. The highest original control changes from 0.696523 to 0.696632. Alongside verification of the consumed inputs and independent score reconstruction, these results show that the original stopping tolerance cannot adequately explain the observed AUC spread.

The controls explain why small random coefficients can rank validation labels well while producing probabilities close to a constant. Removing a score term, repeating randomization and tightening convergence give complementary tests of that explanation; correlated features also contribute. The internal format association and external scheme intervention both expose representation dependence. The 32,695-row validation and 34,593-row group-test analyses do not quantify the HTTPS-attributable fraction of the

64.07-point gain. These checks also do not identify collection-source causality, authenticate historical fits or certify the absence of every leakage path. Threshold-selected performance requires separate evaluation.

Chapter 5—Discussion and Conclusions

5.1 Direct Answers to the Research Questions

CyberSentinel answers Research Questions 1–3 (RQ1–RQ3) through an implemented gateway and controlled experiments. The structural model supplies a substantial internal recall gain; subsequent comparisons investigate its dependencies, available expert corrections and service improvements. Table 4.14 gives the hypothesis decisions. The answers below develop their implications for design.

RQ1—Representation and transfer. Structural features add 64.07 percentage points of internal recall over length alone, with a 95% domain-clustered interval [59.99, 67.63], at false-positive rate (FPR) 0.7747%. Validation labels also follow a format convention whose share of the gain is unquantified. External certified FPR is 90.5487%, versus 9.4513% for length-only; the cascade adds no recall or expert calls (Tables 4.2–4.3). This finding motivates checking the signals behind a feature gain and measuring specificity on the intended source.

The interventions make that test concrete. Scheme spelling changes frozen-model decisions. A separate refit removes scheme sensitivity while reducing true and false positives; its combined fitting and calibration changes require their own labeled evaluation. Paired outputs establish which errors an expert can correct. Shared errors survive every selection between those unchanged decisions.

RQ2—Monitoring and action. The GMM recognizes 115/132 external windows (87.12%), exceeding its 80% target. Its audit alerts are 28/252 windows (11.11%) against 5%, so sensitivity and false alerts must be assessed together. Ordered, stateful replay sends 8,253/8,701 requests (94.8512%) to the expert. The policy retains 69/69 gold

detections and adds 218 certified false positives (Table 4.5). Recognizing a change in inputs does not tell a gateway which response will help. Following alerts into paired decisions supplies that test.

RQ3—Inline tradeoffs. The fixed cascade meets the invocation and primary request-error limits. Pooled p95 is 364.30 ms against a 200 ms budget, and external false-alert safeguards are exceeded. The original 69-pair recall sample gives a corrected lower bound of −6.1533 percentage points under independent labeled-pair assumptions. That bound leaves the −2-point noninferiority margin unresolved; it does not establish inferiority (Tables 4.6–4.7 and 4.14). Measuring these quantities separately reveals where to intervene: detector computation and request handling need distinct experiments, alongside detection quality.

Worker-owned clients improve the synthetic structural-only service; separate scoring tests quantify transformer cost. These experiments connect a proposed change to a measurable consequence. Representation stability is tested with labeled detection, expert selection with corrections, and monitoring with the decisions its alerts change.

5.2 Engineering Findings and Contribution

The benchmark investigation gives URL researchers a concrete check. All 20,209 validation negatives use HTTPS, versus 46.6362% of positives; the negative HTTPS indicator reaches AUC 0.7668. The fixed-weight external intervention also changes decisions. These findings justify checking format conventions before interpreting a gain as transferable detection, without quantifying the HTTPS-attributable fraction. Paired errors establish available expert corrections (Sections 4.8.1 and 4.10.2).

Table 4.21 tests spelling changes with weights fixed. Tables 4.16–4.17 verify exact scheme invariance after refitting and quantify detection changes. Different cohorts and fitting procedures prevent attributing the refit's effect to normalization alone. Together, the internal association and external intervention support a practical benchmark check; collection-source causality remains unresolved.

Routing provides an exact design constraint: 2,261 shared false positives among 2,497 certified negatives impose a 90.5487% floor on unchanged-decision selection. Escalation adds 218 false positives without another gold detection. On the separate cohort, one expert repair is available, but wider gates select none and add 18 false positives. Count available corrections before tuning a gate. Service tests identify a client-path improvement with matching comparable predictions despite changed admission order (Tables 4.18–4.19). Table 4.23 measures expert cost separately.

These experiments make established research questions concrete for a URL gateway. Jitkrittum et al. (2023) motivate examining downstream errors; CyberSentinel counts available corrections, selected repairs and introduced errors. Ahamed et al. (2026) examine external specificity and class-specific transformations; this study connects an internal format association to a fixed-weight scheme effect and a verified invariance property. Monitoring replay follows a change in routing into its outcomes. ExpertFusion's documented calibrated fusion and target-prior adaptation provide architectural context (aI-area, 2026; Hussain et al., 2027). The comparison covers that documented architecture, without claiming full-paper coverage or superiority.

The recall-only analysis supports noninferiority conditional on independent labeled pairs. Every model detects every positive while flagging most negatives; the

original gate calls no expert, and the wider gate adds false positives. Specificity and selected corrections therefore determine the practical value of the preserved recall. Reused observations provide neither independent replication nor guaranteed campaign independence.

5.2.1 Findings and Their Engineering Use

Table 5.1 connects each measured finding to a decision a gateway designer can make.

Table 5.1. Measured engineering findings, supporting evidence and decision implications.

Finding	Measured evidence	Engineering decision and scope
Observed internal gain under benchmark conventions	Internal recall improves by 64.07 percentage points over length alone at 0.7747% FPR (Tables 4.2–4.3).	Treat this as an internal baseline. Validation format–class association qualifies transfer without quantifying the HTTPS-attributable share of group-test recall gain.
Expert correction opportunities and the shared-error bound	Selected internal gate: +34 TP, −40 FP; matched-count control: +5 TP, +9 FP. Separate R cohort: 21 calls, 0 repairs, +18 FP. Original shared-error FPR floor: 90.5487% (Sections 4.8.1, 4.10.2).	Measure available and selected corrections before widening a gate. The internal gain is exploratory; the original shared-error bound applies only to unchanged binary decisions on that cohort.

Finding	Measured evidence	Engineering decision and scope
Measured monitoring sensitivity and response utility	GMM alerts on 115/132 external windows (87.12%); audit false alerts are 11.11%. Escalation adds false positives without gold-recall gain (Table 4.5).	Evaluate recognition of changed inputs and benefit of the subsequent action separately. These are distribution departures, not independently labeled harmful-drift events.
URL-format dependence and verified scheme invariance	Fixed-weight HTTPS intervention: 2,211 fewer FP, 47 fewer gold detections. Separate refit: exact invariance on 8,622 pairs; 251 fewer FP and 184 fewer true-positive detections (Sections 4.8.3, 4.10.1).	Test format dependence and labeled detection together. The refit's 92.90% FPR requires further discrimination work; fixed-weight spelling and refitting use different cohorts.
Recall preservation at a detection ceiling	The original and symmetric cascades have 1,392 concordant detections against the transformer; the conservative envelope is ±0.3143 percentage points (Table 4.22).	The recall-only margin is met conditional on independent labeled pairs. Every model detects every positive while most negatives are flagged; the result establishes no incremental expert utility.

Finding	Measured evidence	Engineering decision and scope
Measured expert cost and client-path improvement	17 expert calls/1,000 inputs: 1.646 times scoring time; p99 6.769 versus 0.226 ms. Separate HTTP comparison: 79.68% median paired p95 reduction with worker clients (Sections 4.9, 4.10.3).	Distinguish detector tails from HTTP client cost. Service evidence is synthetic and structural-only; predictions agree on comparable pairs, while admission-order equivalence remains unmet.
Explained ranking variation in audited shuffled-label controls	All validation negatives use HTTPS, versus 46.6362% of positives. Fixed-score term removal, 99 additional training-label permutations and five stricter continuations support the diagnosis (Section 4.10.4).	Small random fitted directions can rank structured validation labels. Audit fitting inputs, rank behavior and convergence; a near-0.5 score mean cannot validate a control. Historical fits remain unauthenticated.

5.2.2 Validity and Reuse of the Evidence

Appendix A connects inputs, fitted states, outcomes and diagnostics. Input audits, reconstructed scores, permutations and convergence checks support the ranking explanation. Appendix C records information exposure and execution history. This record makes the analyzed fits reproducible; historical-fit authentication and independent replication remain separate requirements.

The shuffled-label investigation explains a puzzling control result. Small residual coefficients can rank format-associated labels well despite nearly constant probabilities. Input audits check the fitted data, term removal measures one contribution, randomization reproduces the spread, and stricter optimization preserves it. Inspecting

coefficients and score ordering alongside mean probability provides a practical test of such controls.

5.2.3 Relationship to the Original Hypotheses

Table 5.2 connects component findings, complete gateway decisions and engineering insight. The internal feature gain, routing bound, scheme invariance and client-path improvement each answer a design question. Joint H1–H3 decisions determine system-level qualification. The contribution is a measured basis for redesign: what to change, why it matters and how to test its consequences.

Table 5.2. Decisions for the complete gateway, component comparisons and diagnostic analyses.

Claim or requirement	**Evidence and achieved components**	**Overall decision**	**What it revealed**
Original H1: representation and low-FPR detection	All internal specificity requirements are met; structural recall exceeds length-only recall on the internal and gold cohorts	Not supported; external specificity and incremental cascade gain do not meet the joint criteria	Check format dependence, specificity and available expert corrections.
Original H2: monitoring and useful escalation	115/132 external windows trigger alerts, meeting the required monitoring sensitivity	Not supported; audit false alarms, policy specificity and incremental gold recall do not meet the joint criteria	Test escalation outcomes separately from alert sensitivity and false alerts.
Original H3: inline detection –service tradeoff	The physical transformer-call and HTTP request-error requirements are met	Not supported; external risk and pooled p95 requirements are not met, and the two-point recall margin is not established	Evaluate detection, recall uncertainty, expert work and HTTP cost together.
D: scheme-neutral detection	Exact invariance on 8,622 paired inputs; 251 fewer false positives and 184 fewer detected positives at frozen operating points	Not supported; recall-gain and low-FPR requirements are not met	Verify scheme stability and labeled detection after refitting.

Claim or requirement	Evidence and achieved components	Overall decision	What it revealed
S: client-topology service comparison	Latency improves: median ratio 0.203249; interval upper 0.213508; worker p95 71.9990 ms; zero worker errors	Not supported as a conjunction; strict agreement is 799/100,000. All 99,999 comparable predictions agree	Client ownership improves service; admission order requires a separate contract.
R: recall-only comparisons	1,392 concordant detections; conservative 95% envelope ±0.3143 points for original and symmetric versus transformer	Both supported conditional on independent labeled pairs; not original H3 or deployment adequacy	Pair ceiling recall with specificity and useful corrections.
R: symmetric benefit and budget	21/2,654 logical calls; 18 added false positives; no added positive detections	Not supported: call limit passes, recall-gain and specificity conditions do not	Low invocation counts need useful selected repairs.
F, C and P diagnostics	Controlled spelling effect; 30 computation arms; five audited controls, followed by 99 additional permutations and five stricter continuations	Representation, computation and control-mechanism findings; no joint diagnostic hypothesis was specified	Separate input sensitivity, computation cost and ranking mechanisms.

5.3 Implications for Practice

Three priorities follow for this gateway. Count available expert repairs, selected repairs and new errors on labeled data from the intended source before widening a gate. Test specificity alongside invariance. Judge monitoring by the decisions its alerts change.

Tranco supplies a popularity-based control with a different evidential role from certified labels (Le Pochat et al., 2019).

Service design requires a measured request path and an explicit response contract. The 99,999 comparable predictions agree; admission order changes and one pair times out. Expert counters and p99 expose costs hidden by invocation fractions. These measurements specify what to preserve and retest when hardware, workloads or arrivals change.

5.4 Limitations and Threats to Interpretation

The results use publisher labels without new forensic adjudication. The 2,497 original certified negatives occupy 234 domains, so the number of URLs exceeds the number of distinct groups. Taking one URL per domain in the expanded cohort changes the estimand. Its narrower recall interval assumes independent labeled pairs and does not account for cross-domain dependence. Neither sample estimates traffic prevalence.

Domain separation leaves shared collection conventions possible. PhiUSIIL lacks the independent per-row provenance needed to fit a noncircular source-only classifier. The controls reproduce a contribution from the HTTPS term and broad ranking variation under randomized training labels on fixed partitions. They do not determine how the convention arose or isolate every feature's contribution. Correlated predictors remain after term removal. The 99 permutations describe algorithmic sensitivity, not independent populations. True validation labels select the five controls' thresholds, and historical fits still lack consumed-input authentication.

The analyst's information changed during the study. The url_norm amendment followed preparation and label-count inspection; checkpoint recovery followed

predictions and partial measurement. Earlier exposure and repeated scoring differ from a blind, single-pass evaluation. Wider gates were selected using exposed internal group-test outcomes, making those gains exploratory, and fixed before the separate domain-balanced comparison. Models and thresholds remain fixed within the stated comparisons (Appendix C). Overlapping GMM windows and activations across domains make saved-outcome resampling conditional on stream order. Label-free probes demonstrate neither semantic preservation nor real-world adversarial robustness.

The operational measurements use one host, a fixed workload order and two original sessions. Session, thermal, cache and elapsed-time effects cannot be separated; closed-loop timing also excludes unsent backlog. Comparison D uses a retrospective publisher-labeled benchmark with shared upstream mechanisms and conflicting collection-month descriptions. Excluding domains does not establish temporal independence. D combines refitting and recalibration with normalization, while its companions test representation behavior rather than live-site equivalence.

The cost experiments measure different workloads. S uses ten primary run pairs of synthetic structural-only HTTP requests. C uses real strings at concurrency one, a designed 1% mixture and five cyclic rotations; its positions are not fully balanced, and it has no run-level confidence interval. Combining them would not establish a transformer-service speedup. Sampled host checks record observed conditions, and saved-evidence calculations verify arithmetic. Neither proves disturbance-free operation or independent replication.

5.5 Future Research, Separate from This Study

Independently adjudicated, time-indexed sources recording formatting separately from outcome would test transferable discrimination and expert corrections. Balanced workload and session order, open-loop arrivals and separate admission-order tests would extend the service evidence.

5.6 Conclusion

CyberSentinel provides a working gateway and controlled design evidence. Structural features add 64.07 percentage points of internal recall. Validation controls reveal format–class association; a fixed-weight intervention establishes external scheme sensitivity. A separate refit removes that sensitivity across 8,622 pairs while reducing true and false positives, distinguishing stability from detection quality.

External-window sensitivity reaches 87.12%; escalation adds 218 false positives without another gold detection. The models' 2,261 shared false positives among 2,497 certified negatives impose an exact 90.5487% floor on unchanged-decision selection. Wider gates on a separate cohort select no repairs and add 18 false positives. Count available corrections before choosing an expert; evaluate the policy by its selections.

Worker-owned clients reduce median paired HTTP p95 by 79.68% in the single-host synthetic structural-only setup. A no-model control supports the client-path explanation; comparable predictions agree despite changed admission order. Expert scoring cost is measured separately. The complete gateway did not meet the prespecified joint criteria of H1–H3; the experiments identify what a redesign must address. CyberSentinel provides the next designer with a working architecture, a measured service improvement, a benchmark check and an exact routing bound.

References

Ahamed, T., Kakon, S. C., Farid, F. A., Uddin, J., & Abdul Karim, H. B. (2026). An integrated evaluation protocol for adversarial robustness, generalization, and explanation stability in URL-based phishing detection. *Frontiers in Computer Science, 8*, Article 1834407. https://doi.org/10.3389/fcomp.2026.1834407

aI-area. (2026). *ExpertFusion* [Source code; inspected repository version]. GitHub. https://github.com/aI-area/ExpertFusion/tree/af739b49a3759e507faeb2f2d6a531ecc6a4dd8e

Alajaji, A. (2026). Classical-first selective cascades for resource-constrained phishing email detection. *Electronics, 15*(14), 3051. https://doi.org/10.3390/electronics15143051

Arp, D., Quiring, E., Pendlebury, F., Warnecke, A., Pierazzi, F., Wressnegger, C., Cavallaro, L., & Rieck, K. (2022). Dos and don'ts of machine learning in computer security. *31st USENIX Security Symposium*, 3971–3988. https://www.usenix.org/conference/usenixsecurity22/presentation/arp

Axelsson, S. (2000). The base-rate fallacy and the difficulty of intrusion detection. *ACM Transactions on Information and System Security, 3*(3), 186–205. https://doi.org/10.1145/357830.357849

Basit, A., Zafar, M., Liu, X., Javed, A. R., Jalil, Z., & Kifayat, K. (2021). A comprehensive survey of AI-enabled phishing attacks detection techniques. *Telecommunication Systems, 76*(1), 139–154. https://doi.org/10.1007/s11235-020-00733-2

Ben-David, S., Blitzer, J., Crammer, K., Kulesza, A., Pereira, F., & Vaughan, J. W. (2010). A theory of learning from different domains. *Machine Learning, 79*(1–2), 151–175. https://doi.org/10.1007/s10994-009-5152-4

Berners-Lee, T., Fielding, R., & Masinter, L. (2005). *Uniform Resource Identifier (URI): Generic syntax* (RFC 3986). RFC Editor. https://doi.org/10.17487/RFC3986

Breiman, L. (2001). Random forests. *Machine Learning, 45*(1), 5–32. https://doi.org/10.1023/A:1010933404324

Chandola, V., Banerjee, A., & Kumar, V. (2009). Anomaly detection: A survey. *ACM Computing Surveys, 41*(3), 1–58. https://doi.org/10.1145/1541880.1541882

Chen, T. Y., Kuo, F.-C., Liu, H., Poon, P.-L., Towey, D., Tse, T. H., & Zhou, Z. Q. (2018). Metamorphic testing: A review of challenges and opportunities. *ACM Computing Surveys, 51*(1), 1–27. https://doi.org/10.1145/3143561

Chicco, D., & Jurman, G. (2020). The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation. *BMC Genomics, 21*(1), Article 6. https://doi.org/10.1186/s12864-019-6413-7

Clopper, C. J., & Pearson, E. S. (1934). The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika, 26*(4), 404–413. https://doi.org/10.1093/biomet/26.4.404

Dean, J., & Barroso, L. A. (2013). The tail at scale. *Communications of the ACM, 56*(2), 74–80. https://doi.org/10.1145/2408776.2408794

Dempster, A. P., Laird, N. M., & Rubin, D. B. (1977). Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society: Series B (Methodological), 39*(1), 1–22. https://doi.org/10.1111/j.2517-6161.1977.tb01600.x

Efron, B. (1979). Bootstrap methods: Another look at the jackknife. *The Annals of Statistics, 7*(1), 1–26. https://doi.org/10.1214/aos/1176344552

Gama, J., Žliobaitė, I., Bifet, A., Pechenizkiy, M., & Bouchachia, A. (2014). A survey on concept drift adaptation. *ACM Computing Surveys, 46*(4), 1–37. https://doi.org/10.1145/2523813

Geifman, Y., & El-Yaniv, R. (2019). SelectiveNet: A deep neural network with an integrated reject option. *Proceedings of Machine Learning Research, 97*, 2151–2159. https://proceedings.mlr.press/v97/geifman19a.html

Gneiting, T., & Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association, 102*(477), 359–378. https://doi.org/10.1198/016214506000001437

Gretton, A., Borgwardt, K. M., Rasch, M. J., Schölkopf, B., & Smola, A. (2012). A kernel two-sample test. *Journal of Machine Learning Research, 13*, 723–773. https://jmlr.org/papers/v13/gretton12a.html

Guo, C., Pleiss, G., Sun, Y., & Weinberger, K. Q. (2017). On calibration of modern neural networks. *Proceedings of Machine Learning Research, 70*, 1321–1330. https://proceedings.mlr.press/v70/guo17a.html

Hannousse, A., & Yahiouche, S. (2020). *Towards benchmark datasets for machine learning based website phishing detection: An experimental study* [Preprint]. arXiv. https://arxiv.org/abs/2010.12847

Hannousse, A., & Yahiouche, S. (2021a). Towards benchmark datasets for machine learning based website phishing detection: An experimental study. *Engineering Applications of Artificial Intelligence, 104*, Article 104347. https://doi.org/10.1016/j.engappai.2021.104347

Hannousse, A., & Yahiouche, S. (2021b). *Web page phishing detection* (Version 3) [Data set]. Mendeley Data. https://doi.org/10.17632/c2gw7fy2j4.3

Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics, 6*(2), 65–70. https://www.jstor.org/stable/4615733

Hussain, M., Abbas, J., Hussain, J., Gu, Y., & Wu, H. (2027). ExpertFusion: Calibrated multi-expert decision fusion for phishing URL detection under distribution shift and target-prior uncertainty. *Expert Systems with Applications, 333*, Article 134089. https://doi.org/10.1016/j.eswa.2026.134089

Jain, A. K., & Gupta, B. B. (2018). Towards detection of phishing websites on client-side using machine learning based approach. *Telecommunication Systems, 68*(4), 687–700. https://doi.org/10.1007/s11235-017-0414-0

Jitkrittum, W., Gupta, N., Menon, A. K., Narasimhan, H., Rawat, A. S., & Kumar, S. (2023). When does confidence-based cascade deferral suffice? *Advances in Neural Information Processing Systems, 36*, 9891–9906. https://doi.org/10.52202/075280-0431

Kalibera, T., & Jones, R. (2013). Rigorous benchmarking in reasonable time. *Proceedings of the 2013 International Symposium on Memory Management*, 63–74. https://doi.org/10.1145/2464157.2464160

Khonji, M., Iraqi, Y., & Jones, A. (2013). Phishing detection: A literature survey. *IEEE Communications Surveys & Tutorials, 15*(4), 2091–2121. https://doi.org/10.1109/SURV.2013.032213.00009

Le, H., Pham, Q., Sahoo, D., & Hoi, S. C. H. (2018). *URLNet: Learning a URL representation with deep learning for malicious URL detection* [Preprint]. arXiv. https://arxiv.org/abs/1802.03162

Le Pochat, V., Van Goethem, T., Tajalizadehkhoob, S., Korczyński, M., & Joosen, W. (2019). Tranco: A research-oriented top sites ranking hardened against manipulation. *Network and Distributed System Security Symposium*. https://doi.org/10.14722/ndss.2019.23386

Li, L., Lin, Y., Chen, D., Ren, S., Li, P., Zhou, J., & Sun, X. (2021). CascadeBERT: Accelerating inference of pre-trained language models via calibrated complete models cascade. *Findings of the Association for Computational Linguistics: EMNLP 2021*, 475–486. https://doi.org/10.18653/v1/2021.findings-emnlp.43

Lipton, Z. C., Wang, Y.-X., & Smola, A. J. (2018). Detecting and correcting for label shift with black box predictors. *Proceedings of Machine Learning Research, 80*, 3122–3130. https://proceedings.mlr.press/v80/lipton18a.html

Loshchilov, I., & Hutter, F. (2019). Decoupled weight decay regularization. *International Conference on Learning Representations*. https://arxiv.org/abs/1711.05101

Lu, J., Liu, A., Dong, F., Gu, F., Gama, J., & Zhang, G. (2019). Learning under concept drift: A review. *IEEE Transactions on Knowledge and Data Engineering, 31*, 2346–2363. https://doi.org/10.1109/TKDE.2018.2876857

Ma, J., Saul, L. K., Savage, S., & Voelker, G. M. (2009). Beyond blacklists: Learning to detect malicious web sites from suspicious URLs. *Proceedings of the 15th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 1245–1254. https://doi.org/10.1145/1557019.1557153

Marchal, S., Armano, G., Grondahl, T., Saari, K., Singh, N., & Asokan, N. (2017). Off-the-Hook: An efficient and usable client-side phishing prevention application. *IEEE Transactions on Computers, 66*(10), 1717–1733. https://doi.org/10.1109/TC.2017.2703808

McNemar, Q. (1947). Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika, 12*(2), 153–157. https://doi.org/10.1007/BF02295996

Newcombe, R. G. (1998). Improved confidence intervals for the difference between binomial proportions based on paired data. *Statistics in Medicine, 17*(22), 2635–2650. https://doi.org/10.1002/(SICI)1097-0258(19981130)17:22%3C2635::AID-SIM954%3E3.0.CO;2-C

Ojala, M., & Garriga, G. C. (2010). Permutation tests for studying classifier performance. *Journal of Machine Learning Research, 11*(62), 1833–1863. https://www.jmlr.org/papers/v11/ojala10a.html

Ovadia, Y., Fertig, E., Ren, J., Nado, Z., Sculley, D., Nowozin, S., Dillon, J. V., Lakshminarayanan, B., & Snoek, J. (2019). *Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift* [Preprint]. arXiv. https://arxiv.org/abs/1906.02530

Pan, S. J., & Yang, Q. (2010). A survey on transfer learning. *IEEE Transactions on Knowledge and Data Engineering, 22*(10), 1345–1359. https://doi.org/10.1109/TKDE.2009.191

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, E. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research, 12*, 2825–2830. https://jmlr.org/papers/v12/pedregosa11a.html

Pendlebury, F., Pierazzi, F., Jordaney, R., Kinder, J., & Cavallaro, L. (2019). TESSERACT: Eliminating experimental bias in malware classification across space and time. *28th USENIX Security Symposium*, 729–746. https://www.usenix.org/conference/usenixsecurity19/presentation/pendlebury

Prasad, A., & Chandra, S. (2024a). PhiUSIIL: A diverse security profile empowered phishing URL detection framework based on similarity index and incremental learning. *Computers & Security, 136*, Article 103545. https://doi.org/10.1016/j.cose.2023.103545

Prasad, A., & Chandra, S. (2024b). *PhiUSIIL Phishing URL (Website)* [Data set]. UCI Machine Learning Repository. https://archive.ics.uci.edu/dataset/967/phiusiil+phishing+url+website+dataset

Rao, R. S., & Pais, A. R. (2019). Detection of phishing websites using an efficient feature-based machine learning framework. *Neural Computing and Applications, 31*(8), 3851–3873. https://doi.org/10.1007/s00521-017-3305-0

Rashid, F., Doyle, B., Han, S. C., & Seneviratne, S. (2024). Phishing URL detection generalisation using unsupervised domain adaptation. *Computer Networks, 245*, Article 110398. https://doi.org/10.1016/j.comnet.2024.110398

Roberts, D. R., Bahn, V., Ciuti, S., Boyce, M. S., Elith, J., Guillera-Arroita, G., Hauenstein, S., Lahoz-Monfort, J. J., Schröder, B., Thuiller, W., Warton, D. I., Wintle, B. A., Hartig, F., & Dormann, C. F. (2017). Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography, 40*(8), 913–929. https://doi.org/10.1111/ecog.02881

Sahingoz, O. K., Buber, E., Demir, O., & Diri, B. (2019). Machine learning based phishing detection from URLs. *Expert Systems with Applications, 117*, 345–357. https://doi.org/10.1016/j.eswa.2018.09.029

Saito, T., & Rehmsmeier, M. (2015). The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE, 10*(3), e0118432. https://doi.org/10.1371/journal.pone.0118432

Schroeder, B., Wierman, A., & Harchol-Balter, M. (2006). Open versus closed: A cautionary tale. *3rd USENIX Symposium on Networked Systems Design & Implementation*, 239–252. https://www.usenix.org/legacy/events/nsdi06/tech/schroeder.html

Schwarz, G. (1978). Estimating the dimension of a model. *The Annals of Statistics, 6*(2), 461–464. https://doi.org/10.1214/aos/1176344136

Sugiyama, M., Krauledat, M., & Müller, K.-R. (2007). Covariate shift adaptation by importance weighted cross validation. *Journal of Machine Learning Research, 8*, 985–1005. https://jmlr.org/papers/v8/sugiyama07a.html

Tallam, K. (2026). *Automated phishing detection for frontier AI inference: Retained research record* (research-record-2026-10-04) [Computer software and data set]. GitHub. https://github.com/KrtiT/automated-phishing-detection-public/releases/tag/research-record-2026-10-04

Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society: Series B (Methodological), 58*(1), 267–288. https://doi.org/10.1111/j.2517-6161.1996.tb02080.x

Tsai, Y.-D., Liow, C., Sheng Siang, Y., & Lin, S.-D. (2024). Toward more generalized malicious URL detection models. *Proceedings of the AAAI Conference on Artificial Intelligence, 38*(19), 21628–21636. https://doi.org/10.1609/aaai.v38i19.30161

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., & Polosukhin, I. (2017). *Attention is all you need* [Preprint]. arXiv. https://arxiv.org/abs/1706.03762

Vu, T. N. (2026a). PhishVN: A time-stamped Vietnamese URL phishing dataset with impersonation-scenario labels and confidence tiers. *Data in Brief, 68*, Article 113195. https://doi.org/10.1016/j.dib.2026.113195

Vu, T. N. (2026b). *PhishVN: A time-stamped Vietnamese URL phishing dataset with impersonation-scenario labels and confidence tiers* (Version 4) [Data set]. Mendeley Data. https://doi.org/10.17632/b97hxbxtpd.4

Yang, L., Guo, W., Hao, Q., Ciptadi, A., Ahmadzadeh, A., Xing, X., & Wang, G. (2021). CADE: Detecting and explaining concept drift samples for security applications. *30th USENIX Security Symposium*, 2327–2344. https://www.usenix.org/conference/usenixsecurity21/presentation/yang-limin

Zhang, X., Zhao, J., & LeCun, Y. (2015). *Character-level convolutional networks for text classification* [Preprint]. arXiv. https://arxiv.org/abs/1509.01626

Appendix A—Evidence and Reproduction Map

The release research-record-2026-10-04, cited as Tallam (2026), contains the original data and measured-code record. Table A.1 uses filenames relative to dissertation/2026-10-04/aggregate-data unless another edition is named. Current original-hypothesis decisions remain in dissertation/2026-10-12/aggregate-data/corrected-primary-gates.csv; primary-gates.csv preserves historical decisions. The October 12 edition supplies frozen-model, routing, computation and audited-control evidence. The October 15 edition adds this manuscript, the retrospective control diagnosis, both frozen diagnostic protocols, per-fit aggregate results and their verification. Earlier editions remain preserved.

Table A.1. Research questions, complete evidence products and reading order.

Question or concern	Read in the manuscript	Aggregate or provenance artifact
Exact commitments and hypotheses	Sections 1.4–1.5	provenance/advisor-deck-scope-crosswalk-20261001.md; current decisions: dissertation/2026-10-12/aggregate-data/corrected-primary-gates.csv
RQ1 representation comparison	Sections 3.6–3.8, 4.2, 5.1	primary-results.json; paired-contrasts.csv; secondary-metrics.csv
RQ2 monitor and routing effects	Sections 3.9–3.11, 4.3, 5.1	external-monitor-windows.csv; external-psi-features.csv; complete-secondary-results.json

Question or concern	Read in the manuscript	Aggregate or provenance artifact
RQ3 service tradeoffs	Sections 3.12, 4.4, 5.1	operational-runs.csv (125 rows); operational-groups.csv (25 rows)
Every primary decision	Section 4.7	dissertation/2026-10-12/aggregate-data/corrected-primary-gates.csv (22 rows); historical ledger: dissertation/2026-10-04/aggregate-data/primary-gates.csv
Ranking, calibration and prevalence	Section 4.5	low-fpr-score-curves.csv; calibration-bins.csv; prevalence-projections.csv
Controls, sources, seeds and probes	Sections 4.5, 5.4	source-contingency.csv; seed-logical-invocations.csv; probe-decisions-and-scores.csv; probe-monitors-and-scores.csv; new control diagnosis: dissertation/2026-10-15/diagnostics/
Integrity and methodological history	Sections 3.13, 4.6	verification.json; secondary-verification.json; provenance/Historical_Evidence_Supplement_2026-10-01.md

The original measured code revision is 77d128377ce5b401437d7179f5cd78fb4294b72c; the bounded follow-up revision is ef8ba5f0b357cf3dd60c4d663e6297d13334460c. The repository's research-archive/2026-10-04 guide supplies download, checksum, materialization and saved-observation recomputation commands. Its catalog inventories licensed raw data, prepared splits, fitted artifacts, predictions, requests, freezes and interrupted attempts. Private execution capabilities and full host/process logs remain excluded or explicitly projected.

Checksums establish identity, and recomputation checks arithmetic; neither constitutes new model fitting, timing or independent empirical replication.

Public method records are in provenance/followup, with detection results in aggregate-data/detection-D and complete service results in aggregate-data/service-S. Materialized follow-up inputs, manifests, model freezes and population admissions are under dissertation/followup-20261001 in the archive. The detection supplement retains two model rows, twenty calibration bins, two invariance rows and domain sizes. Its separate arithmetic implementation checks counts, rank-based AUC, average precision, Brier score and domain-bootstrap summaries. Candidate fitting used Python 3.10.19, NumPy 2.2.6, SciPy 1.15.3 and scikit-learn 1.7.2. Historical analysis-scripts retain controlled-workspace dependencies; the public recomputation adapter and data dictionaries specify the supported use.

The software checks have their own chronology. At the follow-up freeze, 215 focused tests had passed while a broader run was still in progress. That run finished with 13,016 passed, three failed and two skipped. A mock client lacked the async aclose method, which prevented the intended interrupt-signal (SIGINT) injection. A separately retained fixture-only correction passed all six original interruption cases, including three previously passing shift cases. The frozen source, original failure log and initial configuration-selection error before collection remain preserved. Current GitHub Actions results identify the commits they test; they do not change the historical failure record.

Appendix B—Complete Service Comparison Evidence

Table B.1 lists all 80 arms in schedule order. Each includes 1,000 warmups and 10,000 measured attempts. IDs indicate workload, concurrency, pair and client. The structural_detector workload uses the unchanged structural scorer, and no_model provides the response control. The table reports successful-response p95. The machine-readable arm file also includes successful, failed and all-request p50/p95/p99, measured and drain durations, error categories and physical counts. The package retains all eight pooled groups and all 40 pair-agreement rows. An empty failure quantile means there were no failures; it does not mean failures took zero milliseconds.

Table B.1. Complete 80-arm schedule; success-only p95 in milliseconds, attempted throughput in requests/s and terminal error counts.

Arm ID	Success p95	Requests/s	Errors
no_model-c1-pair01-shared	1.1002	1041.13	0
no_model-c1-pair01-worker	1.1114	1037.84	0
no_model-c1-pair02-worker	1.1245	1019.56	0
no_model-c1-pair02-shared	1.0941	1041.69	0
no_model-c1-pair03-shared	1.1086	1035.64	0
no_model-c1-pair03-worker	1.0958	1041.49	0
no_model-c1-pair04-worker	1.1181	1032.21	0

Arm ID	Success p95	Requests/s	Errors
no_model-c1-pair04-shared	1.1015	1040.88	0
no_model-c1-pair05-shared	1.1098	1035.75	0
no_model-c1-pair05-worker	1.0990	1042.91	0
no_model-c1-pair06-worker	1.0963	1040.99	0
no_model-c1-pair06-shared	1.0975	1044.50	0
no_model-c1-pair07-shared	1.0971	1039.30	0
no_model-c1-pair07-worker	1.1045	1040.20	0
no_model-c1-pair08-worker	1.1310	1028.41	0
no_model-c1-pair08-shared	1.1080	1040.75	0
no_model-c1-pair09-shared	1.1011	1036.62	0
no_model-c1-pair09-worker	1.0941	1045.86	0
no_model-c1-pair10-worker	1.1017	1036.23	0
no_model-c1-pair10-shared	1.0979	1042.61	0
no_model-c64-pair01-shared	341.8227	470.09	0
no_model-c64-pair01-worker	66.5180	1652.64	0
no_model-c64-pair02-worker	66.2435	1650.60	0
no_model-c64-pair02-shared	350.9110	466.41	0

Arm ID	Success p95	Requests/s	Errors
no_model-c64-pair03-shared	327.2478	487.55	0
no_model-c64-pair03-worker	65.3162	1653.86	0
no_model-c64-pair04-worker	69.3325	1618.14	0
no_model-c64-pair04-shared	362.3160	440.58	0
no_model-c64-pair05-shared	352.0579	469.31	0
no_model-c64-pair05-worker	67.2167	1647.16	0
no_model-c64-pair06-worker	66.4319	1660.42	0
no_model-c64-pair06-shared	350.3035	461.06	0
no_model-c64-pair07-shared	332.1273	486.86	0
no_model-c64-pair07-worker	67.0119	1671.26	0
no_model-c64-pair08-worker	67.9225	1637.49	0
no_model-c64-pair08-shared	354.3856	466.84	0
no_model-c64-pair09-shared	335.7060	480.74	0
no_model-c64-pair09-worker	68.1728	1640.31	0
no_model-c64-pair10-worker	66.3344	1645.62	0
no_model-c64-pair10-shared	350.3558	466.32	0
structural_detector-c1-pair01-shared	1.4180	819.89	0

Arm ID	Success p95	Requests/s	Errors
structural_detector-c1-pair01-worker	1.4179	821.63	0
structural_detector-c1-pair02-worker	1.4262	819.52	0
structural_detector-c1-pair02-shared	1.4287	821.01	0
structural_detector-c1-pair03-shared	1.4133	826.39	0
structural_detector-c1-pair03-worker	1.4262	824.31	0
structural_detector-c1-pair04-worker	1.4413	816.86	0
structural_detector-c1-pair04-shared	1.4103	826.39	0
structural_detector-c1-pair05-shared	1.4393	814.60	0
structural_detector-c1-pair05-worker	1.4103	819.61	0
structural_detector-c1-pair06-worker	1.4283	819.98	0
structural_detector-c1-pair06-shared	1.4164	822.56	0
structural_detector-c1-pair07-shared	1.4487	816.54	0
structural_detector-c1-pair07-worker	1.4211	821.83	0
structural_detector-c1-pair08-worker	1.4415	817.97	0
structural_detector-c1-pair08-shared	1.4195	820.35	0
structural_detector-c1-pair09-shared	1.4238	814.64	0
structural_detector-c1-pair09-worker	1.4304	817.64	0

Arm ID	Success p95	Requests/s	Errors
structural_detector-c1-pair10-worker	1.4218	818.32	0
structural_detector-c1-pair10-shared	1.4200	822.27	0
structural_detector-c64-pair01-shared	343.1552	467.91	0
structural_detector-c64-pair01-worker	72.6055	1517.21	0
structural_detector-c64-pair02-worker	67.2667	1575.63	0
structural_detector-c64-pair02-shared	371.7900	442.31	1
structural_detector-c64-pair03-shared	359.2445	456.39	0
structural_detector-c64-pair03-worker	70.0094	1533.45	0
structural_detector-c64-pair04-worker	72.8568	1509.99	0
structural_detector-c64-pair04-shared	358.8575	450.21	0
structural_detector-c64-pair05-shared	397.6750	416.67	0
structural_detector-c64-pair05-worker	71.5949	1532.53	0
structural_detector-c64-pair06-worker	74.7204	1503.85	0
structural_detector-c64-pair06-shared	342.2972	472.64	0
structural_detector-c64-pair07-shared	362.0759	442.19	0
structural_detector-c64-pair07-worker	74.3130	1525.00	0
structural_detector-c64-pair08-worker	71.7555	1522.63	0

Arm ID	Success p95	Requests/s	Errors
structural_detector-c64-pair08-shared	352.6524	464.16	0
structural_detector-c64-pair09-shared	341.7835	475.67	0
structural_detector-c64-pair09-worker	73.6319	1509.45	0
structural_detector-c64-pair10-worker	71.2943	1539.79	0
structural_detector-c64-pair10-shared	353.7333	459.72	0

The service uses the unchanged ef8ba5f0b357cf3dd60c4d663e6297d13334460c checkout. Recovery authorization binds the separate launcher, tests, amendment, original manifest and receipt preserving the interrupted attempt. A separately implemented arithmetic verifier checks sorted linear quantiles, medians, paired responses, physical counts and the fixed PCG64 bootstrap. Its verification record identifies the retained sources and five aggregate CSVs. This is a second calculation rather than an external replication. The original power interruption, shared-client timeout and unmet strict-agreement requirement all remain in the evidence. Document synthesis performs no fitting, prediction or measurement.

Appendix C—Methodological Chronology and Execution Record

The August 20 advisor-deck working thesis (slide 5) stated: “CyberSentinel will test low-FPR structural and character-level detection, GMM-guided escalation, and whether selective execution stays within two percentage points of transformer-only recall while invoking the transformer no more than 30%, keeping p95 at or below 200 ms, and holding errors below 0.1%.”

Protocol v1.10 governs the research record. The accepted structural and character artifacts were selected and validated before evaluation. A failed liblinear predecessor led to the documented SAGA specification, while later scoring and durable-publication corrections remain separately identified. Group-test text displays exposed data to the analyst before scoring. The September 21 singleton-scoring amendment followed a development comparison that changed one transformer decision. Historically selected operating points were retained without recalibration.

Initial external parsing admitted no gold or Tranco population because raw publisher strings lacked schemes. The amendment adopted exact publisher url_norm for both parsing and model input while preserving raw cells. Preparation and inspection of label counts occurred before this change; amended predictions occurred afterward. Source mappings, overlap rules, models and hypothesis thresholds remained unchanged.

Four interrupted attempts preceded the accepted operational completion. The September 30 directive authorized bounded recovery after predictions and partial measurement. An all-or-none review admitted cells 1–72 from the retained prefix, and the continuation measured cells 73–125 in order. Cell 73 was measured afresh rather than assembled from attempts. The original primary H3 cells remain in the first session, while

descriptive group 71–75 crosses the session boundary. Every failed predecessor is kept separate from accepted evidence.

D changes the representation, refits and recalibrates using the original development partitions, then evaluates the admitted retrospective benchmark. A preceding schedule ended at an environmental stop condition during S. The disclosed replacement rule preserved those nonprimary observations, required a complete new 80-arm schedule and prohibited pooling. Final S retains the unchanged structural scorer and every original response requirement, including the strict-equivalence condition that was not met.

After the October 12 review, corrected paired inference changed one original component decision. P, F, R and C were then specified and executed once: five audited shuffled-label fits, frozen-model scheme interventions, domain-balanced retrospective routing comparisons and 30 actual-computation arms. Frozen research commit: f77027b3963f4e7ee6b831b9f09ff04c71491001. Execution-profile SHA-256: 84e4d47f2c7c6249766dc677b6b7c2cad5be8fdaefb65195d786be418463c4d6. Independent saved-evidence verification checks 506 bound files and 258 recorded host observations, without new fitting or prediction. Host checks cover sampled power, thermal and known workload indicators, not every operating-system activity.

The original manuscripts, failures, source evidence and requirements remain preserved. Extended_Methods_and_Chronology_2026-10-12.docx gives the detailed history; Section 3.8 specifies corrected inference. File identities establish provenance without establishing access or replication. Public aggregates support examination. New raw URLs, domains, record identifiers and record-level predictions remain private.

The control-mechanism analysis uses only the original training and validation partitions. Its protocol retains five saved P models, a fixed set of 99 additional label permutations and five stricter numerical continuations. The public dissertation/2026-10-15 directory contains protocols, source, aggregate coefficients and metrics, a diagnostic report and independent metric-verification results. Reproduction instructions distinguish aggregate recalculation from runs requiring original partitions and restricted fit receipts. Cache matrices, URLs, domains and record identifiers remain excluded. The hypothesis criteria and decisions are unchanged; the retained protocols document the diagnostic analyses separately from the original hypothesis tests.